

Credit Risk Prediction using Explainable Machine Learning (Interpretable vs Black-Box Models)

A Comparative Analysis using SHAP and LIME

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Dataset: Home Credit Default Risk (Kaggle)

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Set up & Imports

```
In [1]: import subprocess
import sys

required = ['xgboost', 'lightgbm', 'shap', 'lime']
for package in required:
    try:
        __import__(package)
    except ImportError:
        print(f"Installing {package}...")
        subprocess.check_call([sys.executable, '-m', 'pip', 'install', package])
        print(f"{package} installed")

from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from lightgbm import LGBMClassifier
from sklearn.metrics import (
    roc_auc_score, f1_score,
    roc_curve, classification_report
)
from scipy.stats import ks_2samp
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')

# metric helper
def compute_metrics(model_name, y_true, y_prob, y_pred):
    auc = roc_auc_score(y_true, y_prob)
    gini = 2 * auc - 1
    default_probs = y_prob[y_true == 1]
    nondefault_probs = y_prob[y_true == 0]
    ks_stat, _ = ks_2samp(default_probs, nondefault_probs)
    f1 = f1_score(y_true, y_pred)
```

```

print(f"\n{'='*45}")
print(f"  {model_name}")
print(f"{'='*45}")
print(f"  AUC-ROC : {auc:.4f}")
print(f"  Gini    : {gini:.4f}")
print(f"  KS Stat : {ks_stat:.4f}")
print(f"  F1 Score: {f1:.4f}")
return {
    'Model' : model_name,
    'AUC-ROC' : round(auc, 4),
    'Gini' : round(gini, 4),
    'KS Stat' : round(ks_stat, 4),
    'F1 Score': round(f1, 4)
}

results = []
print("\nSetup complete – ready to train models")

```

Setup complete – ready to train models

```

In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings

from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.impute import SimpleImputer

warnings.filterwarnings('ignore')

# plot styling
plt.rcParams['figure.figsize'] = (12, 5)
plt.rcParams['figure.dpi'] = 100
plt.rcParams['axes.spines.top'] = False
plt.rcParams['axes.spines.right'] = False

# display settings
pd.set_option('display.max_columns', 100)
pd.set_option('display.float_format', '{:.4f}'.format)

print("Libraries loaded successfully")

```

Libraries loaded successfully

```

In [3]: df = pd.read_csv('/Users/gemkrairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRis

print(f"Rows:    {df.shape[0]:,}")
print(f"Columns: {df.shape[1]}")

```

Rows: 307,511
Columns: 122

```

In [4]: import os
print(os.getcwd())

```

/Users/gemkrairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML

Section 1: Exploratory Data Analysis (EDA)

Dataset: Home Credit Default Risk - 'application_train.csv'

Source: Home Credit Group via Kaggle

Citation: Home Credit Group (2018). *Home Credit Default Risk*[Dataset]. Kaggle.
<https://www.kaggle.com/c/home-credit-default-risk>

Coverage:

- 1. Load and inspect the raw dataset.
- 2. Check the target variables distribution (numeric vs categorical)
- 3. Identify feature types (numeric vs categorical)
- 4. Analyse missing values per column
- 5. Examine correlations with target variable
- 6. Produce EDA Summary

Key Decisions:

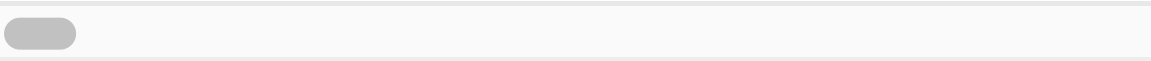
- Columns with >40% missing values will be dropped in preprocessing.
- EXT_SOURCE_1 retained despite high numbers of missing values (credit bureau score)
- Class imbalance (~8%) confirms need for class weighting strategy.

In [5]: df.head()

Out [5]:

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CA
0	100002	1	Cash loans	M	
1	100003	0	Cash loans	F	
2	100004	0	Revolving loans	M	
3	100006	0	Cash loans	F	
4	100007	0	Cash loans	M	

5 rows × 122 columns



In [6]: df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 307511 entries, 0 to 307510
Columns: 122 entries, SK_ID_CURR to AMT_REQ_CREDIT_BUREAU_YEAR
dtypes: float64(65), int64(41), object(16)
memory usage: 286.2+ MB
```

In [7]: df.describe()

Out [7]:

	SK_ID_CURR	TARGET	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CR
count	307511.0000	307511.0000	307511.0000	307511.0000	307511.0
mean	278180.5186	0.0807	0.4171	168797.9193	599025.9
std	102790.1753	0.2724	0.7221	237123.1463	402490.7
min	100002.0000	0.0000	0.0000	25650.0000	45000.0
25%	189145.5000	0.0000	0.0000	112500.0000	270000.0
50%	278202.0000	0.0000	0.0000	147150.0000	513531.0
75%	367142.5000	0.0000	1.0000	202500.0000	808650.0
max	456255.0000	1.0000	19.0000	117000000.0000	4050000.0

8 rows × 106 columns

```
In [8]: print("Target distribution:")
print(df['TARGET'].value_counts())
print(f"\nDefault rate: {df['TARGET'].mean():.2%}")
print(f"Non-default: {(df['TARGET'] == 0).sum():,}")
print(f"Default:      {(df['TARGET'] == 1).sum():,}")
```

Target distribution:
 TARGET
 0 282686
 1 24825
 Name: count, dtype: int64

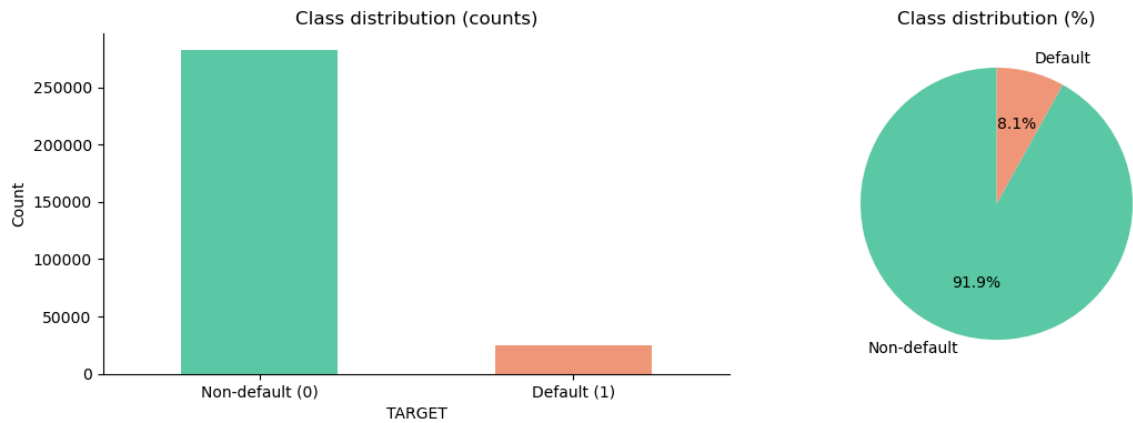
Default rate: 8.07%
 Non-default: 282,686
 Default: 24,825

```
In [9]: fig, axes = plt.subplots(1, 2, figsize=(12, 4))

df['TARGET'].value_counts().plot(
    kind='bar', ax=axes[0],
    color=['#5DCAA5', '#F0997B'],
    edgecolor='none'
)
axes[0].set_title('Class distribution (counts)')
axes[0].set_xticklabels(['Non-default (0)', 'Default (1)'], rotation=0)
axes[0].set_ylabel('Count')

axes[1].pie(
    df['TARGET'].value_counts(),
    labels=['Non-default', 'Default'],
    autopct='%1.1f%%',
    colors=['#5DCAA5', '#F0997B'],
    startangle=90
)
axes[1].set_title('Class distribution (%)')

plt.tight_layout()
plt.show()
```



```
In [10]: missing = pd.DataFrame({
    'missing_count': df.isnull().sum(),
    'missing_pct': (df.isnull().sum() / len(df) * 100).round(2)
}).sort_values('missing_pct', ascending=False)

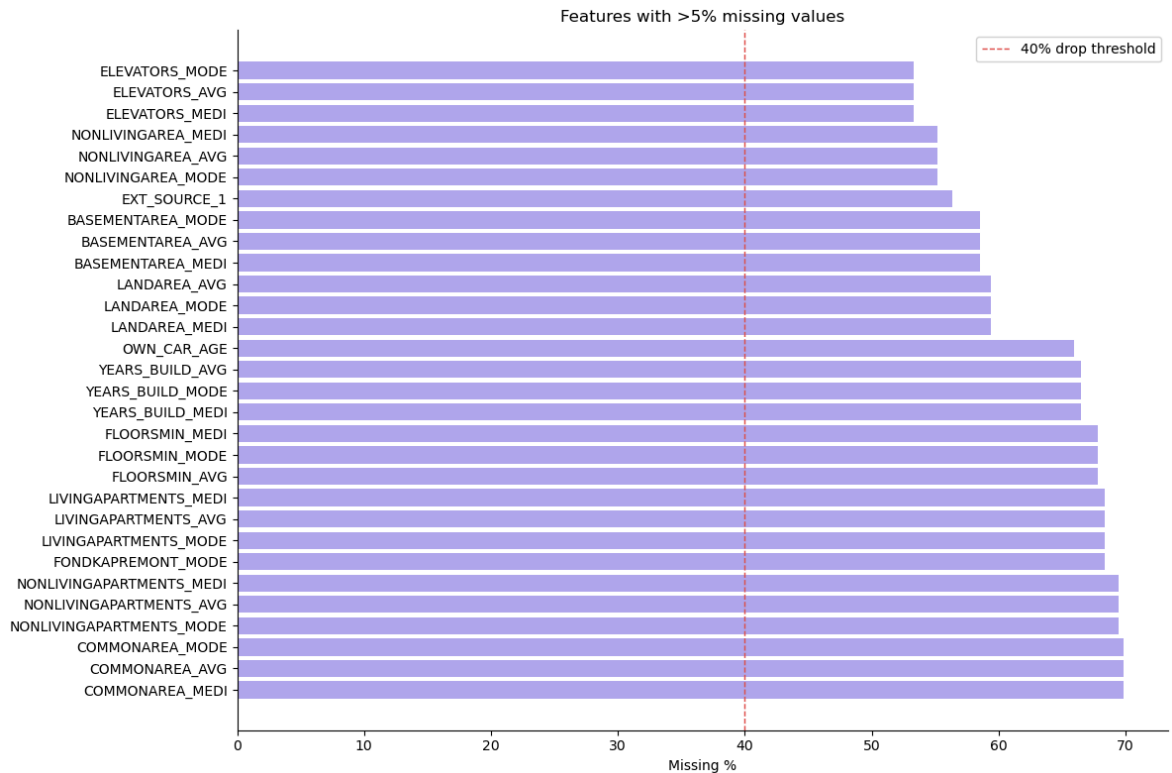
missing_cols = missing[missing['missing_count'] > 0]
print(f"Columns with missing values: {len(missing_cols)}")
print(missing_cols.head(20))
```

Columns with missing values: 67

	missing_count	missing_pct
COMMONAREA_MEDI	214865	69.8700
COMMONAREA_AVG	214865	69.8700
COMMONAREA_MODE	214865	69.8700
NONLIVINGAPARTMENTS_MODE	213514	69.4300
NONLIVINGAPARTMENTS_AVG	213514	69.4300
NONLIVINGAPARTMENTS_MEDI	213514	69.4300
FONDKAPREMONT_MODE	210295	68.3900
LIVINGAPARTMENTS_MODE	210199	68.3500
LIVINGAPARTMENTS_AVG	210199	68.3500
LIVINGAPARTMENTS_MEDI	210199	68.3500
FLOORSMIN_AVG	208642	67.8500
FLOORSMIN_MODE	208642	67.8500
FLOORSMIN_MEDI	208642	67.8500
YEARS_BUILD_MEDI	204488	66.5000
YEARS_BUILD_MODE	204488	66.5000
YEARS_BUILD_AVG	204488	66.5000
OWN_CAR_AGE	202929	65.9900
LANDAREA_MEDI	182590	59.3800
LANDAREA_MODE	182590	59.3800
LANDAREA_AVG	182590	59.3800

```
In [11]: # columns with more than 5% missing
to_plot = missing[missing['missing_pct'] > 5].head(30)

fig, ax = plt.subplots(figsize=(12, 8))
ax.barh(to_plot.index, to_plot['missing_pct'], color='#AFA9EC', edgecolor='black')
ax.axvline(x=40, color='#E24B4A', linestyle='--', linewidth=1, label='40%')
ax.set_xlabel('Missing %')
ax.set_title('Features with >5% missing values')
ax.legend()
plt.tight_layout()
plt.show()
```



```
In [12]: # correlation with target
exclude = ['SK_ID_CURR', 'TARGET']
numeric_cols = df.select_dtypes(include=[np.number]).columns.tolist()
numeric_cols = [c for c in numeric_cols if c not in exclude]

corr = df[numeric_cols + ['TARGET']].corr()['TARGET'].drop('TARGET')
corr_abs = corr.abs().sort_values(ascending=False)

print("Top 15 features correlated with TARGET:")
print(corr_abs.head(15).round(4))
```

Top 15 features correlated with TARGET:

EXT_SOURCE_3	0.1789
EXT_SOURCE_2	0.1605
EXT_SOURCE_1	0.1553
DAYS_BIRTH	0.0782
REGION_RATING_CLIENT_W_CITY	0.0609
REGION_RATING_CLIENT	0.0589
DAYS_LAST_PHONE_CHANGE	0.0552
DAYS_ID_PUBLISH	0.0515
REG_CITY_NOT_WORK_CITY	0.0510
FLAG_EMP_PHONE	0.0460
DAYS_EMPLOYED	0.0449
REG_CITY_NOT_LIVE_CITY	0.0444
FLAG_DOCUMENT_3	0.0443
FLOORSMAX_AVG	0.0440
FLOORSMAX_MEDI	0.0438

Name: TARGET, dtype: float64

```
In [13]: exclude = ['SK_ID_CURR', 'TARGET']
numeric_cols = df.select_dtypes(include=[np.number]).columns.tolist()
numeric_cols = [c for c in numeric_cols if c not in exclude]
categorical_cols = df.select_dtypes(include=['object']).columns.tolist()

missing = pd.DataFrame({
    'missing_count': df.isnull().sum(),
```

```

    'missing_pct': (df.isnull().sum() / len(df) * 100).round(2)
}).sort_values('missing_pct', ascending=False)
drop_threshold = 40
drop_cols = missing[missing['missing_pct'] > drop_threshold].index.tolist

print("=" * 50)
print("EDA SUMMARY – copy this into your methodology")
print("=" * 50)
print(f"Total rows:                {len(df):,}")
print(f"Total columns:              {df.shape[1]}")
print(f"Numeric features:            {len(numeric_cols)}")
print(f"Categorical features:       {len(categorical_cols)}")
print(f"Default rate:                {df['TARGET'].mean():.2%}")
print(f"Columns >40% missing:       {len(drop_cols)}")
print(f"Columns to drop:             {drop_cols}")
print("=" * 50)

```

```

=====
EDA SUMMARY – copy this into your methodology
=====

```

```

Total rows:                307,511
Total columns:             122
Numeric features:          104
Categorical features:       16
Default rate:              8.07%
Columns >40% missing:      49
Columns to drop:           ['COMMONAREA_MEDI', 'COMMONAREA_AVG', 'COMMONAR
EA_MODE', 'NONLIVINGAPARTMENTS_MODE', 'NONLIVINGAPARTMENTS_AVG', 'NONLIVIN
GAPARTMENTS_MEDI', 'FONDKAPREMONT_MODE', 'LIVINGAPARTMENTS_MODE', 'LIVINGA
PARTMENTS_AVG', 'LIVINGAPARTMENTS_MEDI', 'FLOORSMIN_AVG', 'FLOORSMIN_MOD
E', 'FLOORSMIN_MEDI', 'YEARS_BUILD_MEDI', 'YEARS_BUILD_MODE', 'YEARS_BUILD
_AVG', 'OWN_CAR_AGE', 'LANDAREA_MEDI', 'LANDAREA_MODE', 'LANDAREA_AVG', 'B
ASEMENTAREA_MEDI', 'BASEMENTAREA_AVG', 'BASEMENTAREA_MODE', 'EXT_SOURCE_
1', 'NONLIVINGAREA_MODE', 'NONLIVINGAREA_AVG', 'NONLIVINGAREA_MEDI', 'ELEV
ATORS_MEDI', 'ELEVATORS_AVG', 'ELEVATORS_MODE', 'WALLSMATERIAL_MODE', 'APA
RTMENTS_MEDI', 'APARTMENTS_AVG', 'APARTMENTS_MODE', 'ENTRANCES_MEDI', 'ENT
RANCES_AVG', 'ENTRANCES_MODE', 'LIVINGAREA_AVG', 'LIVINGAREA_MODE', 'LIVIN
GAREA_MEDI', 'HOUSETYPE_MODE', 'FLOORSMAX_MODE', 'FLOORSMAX_MEDI', 'FLOORS
MAX_AVG', 'YEARS_BEGINEXPLUATATION_MODE', 'YEARS_BEGINEXPLUATATION_MEDI',
'YEARS_BEGINEXPLUATATION_AVG', 'TOTALAREA_MODE', 'EMERGENCYSTATE_MODE']
=====

```

EDA Findings Summary

The EDA reveals the following key characteristics of the dataset:

Property	Value
Total rows	307,511
Total columns (raw)	122
Numeric features	104
Categorical features	16
Default rate (TARGET=1)	8.07%
Columns with >40% missing	49

Implications: The 8.07% default rate confirms moderate class imbalance. The majority class (non-default) outnumber the minority class (default) at approximately 11:1. This will be addressed in preprocessing using 'class_weight='balanced'' rather than synthetic oversampling (SMOTE) to preserve data integrity.

Columns to drop: 49 features exceed the 40% missing threshold. EXT_SOURCE_1- is an exception - retained and imputed due to its established importance in credit risk literature.

Section 2: PREPROCESSING

This section prepares the data for modelling through a series of documented decisions. Every decision here is justified in the methodology chapter of the thesis.

Preprocessing pipeline:

Step	Method	Justification
Drop high-missing columns	Threshold: >40% missing	Retains informative features only
Drop ID column	Remove SK_ID_CURR	No predictive signal
Label encode categoricals	LabelEncoder	Compatible with all 6 models
Impute missing values	Median substitution	Robust to skewed distributions
Handle class imbalance	class_weight='balanced'	Preserves original data distribution
Scale numeric features	StandardScaler (fit on train only)	Required for logistic regression
Train/val/test split	70/15/15, stratified	Preserves default rate across splits

Note:

- Scaler is fit on training data only - applying it to val/test prevents data leakage.
- Test set is held out - touched exactly once for final evaluation.
- Random seed = 42 - all splits are reproducible.

Steps:

1. Drop high-missing columns (>40%)
2. Separate features and target
3. Encode categorical variables
4. Impute missing values
5. Train / Validation / Test split (70/15/15)
6. Save processed splits Drop high missing columns


```
In [14]: # recalculate missing
missing = pd.DataFrame({
    'missing_pct': (df.isnull().sum() / len(df) * 100)
})

drop_cols = missing[missing['missing_pct'] > 40].index.tolist()

# rescue EXT_SOURCE_1
drop_cols = [col for col in drop_cols if col != 'EXT_SOURCE_1']

# drop ID column and high missing columns
df_clean = df.drop(columns=['SK_ID_CURR'] + drop_cols)

print(f"Remaining shape: {df_clean.shape}")
print(f"Dropped {len(drop_cols)} columns")
```

Remaining shape: (307511, 73)
Dropped 48 columns

```
In [15]: X = df_clean.drop(columns=['TARGET'])
y = df_clean['TARGET']

print(f"Features: {X.shape}")
print(f"Target: {y.shape}")
print(f"Default rate: {y.mean():.2%}")
```

Features: (307511, 72)
Target: (307511,)
Default rate: 8.07%

```
In [16]: cat_cols = X.select_dtypes(include='object').columns.tolist()
print(f"Categorical columns to encode: {cat_cols}")

le = LabelEncoder()
for col in cat_cols:
    X[col] = le.fit_transform(X[col].astype(str))

print("Encoding complete")
```

Categorical columns to encode: ['NAME_CONTRACT_TYPE', 'CODE_GENDER', 'FLAG_OWN_CAR', 'FLAG_OWN_REALTY', 'NAME_TYPE_SUITE', 'NAME_INCOME_TYPE', 'NAME_EDUCATION_TYPE', 'NAME_FAMILY_STATUS', 'NAME_HOUSING_TYPE', 'OCCUPATION_TYPE', 'WEEKDAY_APPR_PROCESS_START', 'ORGANIZATION_TYPE']
Encoding complete

```
In [17]: imputer = SimpleImputer(strategy='median')
X_imputed = pd.DataFrame(
    imputer.fit_transform(X),
    columns=X.columns
)

print(f"Missing values remaining: {X_imputed.isnull().sum().sum()}")
print("Imputation complete")
```

Missing values remaining: 0
Imputation complete

```
In [18]: # split off test set first (15%)
X_temp, X_test, y_temp, y_test = train_test_split(
    X_imputed, y,
    test_size=0.15,
```

```

    random_state=42,
    stratify=y
)

# split remaining into train (70%) and validation (15%)
X_train, X_val, y_train, y_val = train_test_split(
    X_temp, y_temp,
    test_size=0.1765,
    random_state=42,
    stratify=y_temp
)

print(f"Train:      {X_train.shape[0]:,} rows ({y_train.mean():.2%} default)")
print(f"Validation: {X_val.shape[0]:,} rows ({y_val.mean():.2%} default)")
print(f"Test:        {X_test.shape[0]:,} rows ({y_test.mean():.2%} default)")

```

```

Train:      215,249 rows (8.07% default)
Validation: 46,135 rows (8.07% default)
Test:       46,127 rows (8.07% default)

```

In [19]: **import** os

```

# create the processed folder if it does not exist
os.makedirs('/Users/gemkairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML/')

# save all splits
X_train.to_csv('/Users/gemkairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML/X_train.csv')
X_val.to_csv('/Users/gemkairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML/X_val.csv')
X_test.to_csv('/Users/gemkairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML/X_test.csv')
y_train.to_csv('/Users/gemkairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML/y_train.csv')
y_val.to_csv('/Users/gemkairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML/y_val.csv')
y_test.to_csv('/Users/gemkairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML/y_test.csv')

print("Saved successfully:")
print(f" X_train: {X_train.shape}")
print(f" X_val:   {X_val.shape}")
print(f" X_test:  {X_test.shape}")
print(f" y_train: {y_train.shape}")
print(f" y_val:   {y_val.shape}")
print(f" y_test:  {y_test.shape}")

```

```

Saved successfully:
X_train: (215249, 72)
X_val:   (46135, 72)
X_test:  (46127, 72)
y_train: (215249,)
y_val:   (46135,)
y_test:  (46127,)

```

In [20]: **import** os

```

files = os.listdir('/Users/gemkairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML/')
print("Files in processed folder:")
for f in sorted(files):
    print(f" {f}")

```

Files in processed folder:

```
X_test.csv
X_train.csv
X_val.csv
lime_fidelity_scores.csv
lime_importance_lgbm.csv
lime_importance_rf.csv
lime_importance_xgb.csv
lime_stability_scores.csv
scaler.pkl
shap_importance_lgbm.csv
shap_importance_rf.csv
shap_importance_xgb.csv
shap_lime_spearman_table.csv
shap_top10_comparison.csv
shap_values_lgbm.npy
shap_values_rf.npy
shap_values_xgb.npy
test_predictions.csv
y_test.csv
y_train.csv
y_val.csv
```

In [21]: `from sklearn.preprocessing import StandardScaler`

```
scaler = StandardScaler()

# fit on training set only
X_train_scaled = pd.DataFrame(
    scaler.fit_transform(X_train),
    columns=X_train.columns
)

# apply same scaler to validation and test
X_val_scaled = pd.DataFrame(
    scaler.transform(X_val),
    columns=X_val.columns
)

X_test_scaled = pd.DataFrame(
    scaler.transform(X_test),
    columns=X_test.columns
)

print("Scaling complete")
print(f"X_train_scaled mean (should be ~0): {X_train_scaled.mean().mean()}")
print(f"X_train_scaled std (should be ~1): {X_train_scaled.std().mean()}")
```

Scaling complete

X_train_scaled mean (should be ~0): -0.0000

X_train_scaled std (should be ~1): 0.9861

In [22]:

```
print("=== PREPROCESSING VERIFICATION ===")
print(f"\nShapes:")
print(f" X_train: {X_train.shape} | y_train: {y_train.shape}")
print(f" X_val: {X_val.shape} | y_val: {y_val.shape}")
print(f" X_test: {X_test.shape} | y_test: {y_test.shape}")

print(f"\nDefault rates (should all be ~8.07%):")
print(f" Train: {y_train.mean():.2%}")
```

```
print(f" Validation: {y_val.mean():.2%}")
print(f" Test:      {y_test.mean():.2%}")

print(f"\nMissing values remaining:")
print(f" X_train: {X_train.isnull().sum().sum()}")
print(f" X_val:   {X_val.isnull().sum().sum()}")
print(f" X_test:  {X_test.isnull().sum().sum()}")

print(f"\nScaling check:")
print(f" X_train_scaled mean: {X_train_scaled.mean().mean():.4f} (should")
print(f" X_train_scaled std:  {X_train_scaled.std().mean():.4f} (should")

print(f"\nFeature count: {X_train.shape[1]} features")
print("\n=== ALL CHECKS PASSED ===" if X_train.isnull().sum().sum() == 0
```

=== PREPROCESSING VERIFICATION ===

Shapes:

```
X_train: (215249, 72) | y_train: (215249,)
X_val:   (46135, 72) | y_val:   (46135,)
X_test:  (46127, 72) | y_test:  (46127,)
```

Default rates (should all be ~8.07%):

```
Train:      8.07%
Validation: 8.07%
Test:       8.07%
```

Missing values remaining:

```
X_train: 0
X_val:   0
X_test:  0
```

Scaling check:

```
X_train_scaled mean: -0.0000 (should be ~0)
X_train_scaled std:  0.9861 (should be ~1)
```

Feature count: 72 features

=== ALL CHECKS PASSED ===

Preprocessing Verification - All Checks Passed.

Check	Result
Train shape	(215,249, 72)
Validation shape	(46,135, 72)
Test shape	(46,127, 72)
Missing values (all splits)	0
Default rate — train	8.07%
Default rate — validation	8.07%
Default rate — test	8.07%
Scaler mean (should be ~0)	-0.0000
Scaler std (should be ~1)	0.9861

Check	Result
Feature count	72

Stratification confirmed: All three splits show identical default rates, confirming stratified sampling worked correctly.

No data leakage: Scaler fitted on training data only. Test set not used until final evaluation.

Saving scaler object

```
In [23]: import pickle

with open('/Users/gemkrairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML/ar
pickle.dump(scaler, f)

print("Scaler saved")
```

Scaler saved

Section 2.1 Feature Selection

Applying feature selection before model training to:

1. Remove low-importance or redundant features
2. Improve model interpretability
3. Enable comparison between full feature set and reduced set

Method: SelectKBest using f_classif (ANOVA F-statistic) comparing top 20, 30, and 40 features against full 72 features.

```
In [59]: from sklearn.feature_selection import SelectKBest, f_classif, mutual_info
import matplotlib.pyplot as plt

print("Running feature selection...")

# method 1 - ANOVA F-statistic
selector_f = SelectKBest(f_classif, k='all')
selector_f.fit(X_train, y_train)

# method 2 - mutual information
selector_mi = SelectKBest(mutual_info_classif, k='all')
selector_mi.fit(X_train, y_train)

# create importance dataframe
feature_scores = pd.DataFrame({
    'Feature' : X_train.columns,
    'F_Score' : selector_f.scores_,
    'MI_Score' : selector_mi.scores_
}).sort_values('F_Score', ascending=False).reset_index(drop=True)

feature_scores['F_Rank'] = feature_scores.index + 1
feature_scores['MI_Rank'] = feature_scores['MI_Score'].rank(ascending=False)
```

```
print("Top 20 features by ANOVA F-score:")
print(feature_scores.head(20)[['Feature', 'F_Score', 'MI_Score']].to_stri
```

The history saving thread hit an unexpected error (OperationalError('attempt to write a readonly database')).History will not be written to the database.

Running feature selection...

Top 20 features by ANOVA F-score:

	Feature	F_Score	MI_Score
	EXT_SOURCE_2	5428.2774	0.0119
	EXT_SOURCE_3	5343.4073	0.0154
	EXT_SOURCE_1	2095.1776	0.0155
	DAYS_BIRTH	1336.8549	0.0040
REGION_RATING_CLIENT_W_CITY		810.8316	0.0204
REGION_RATING_CLIENT		737.2141	0.0186
NAME_EDUCATION_TYPE		597.8033	0.0274
DAYS_LAST_PHONE_CHANGE		596.1505	0.0023
CODE_GENDER		593.5204	0.0146
REG_CITY_NOT_WORK_CITY		560.5254	0.0075
DAYS_ID_PUBLISH		532.3734	0.0005
FLAG_DOCUMENT_3		462.1705	0.0392
FLAG_EMP_PHONE		457.4465	0.0392
DAYS_EMPLOYED		436.9955	0.0079
REG_CITY_NOT_LIVE_CITY		434.1818	0.0020
NAME_INCOME_TYPE		420.6836	0.0201
DAYS_REGISTRATION		348.0243	0.0003
AMT_GOODS_PRICE		340.1084	0.0058
REGION_POPULATION_RELATIVE		306.6639	0.0041
DEF_60_CNT_SOCIAL_CIRCLE		253.9908	0.0009

```
In [60]: #Select top features and compare
# compare model performance across different feature counts
from sklearn.linear_model import LogisticRegression

k_values = [20, 30, 40, 72] # 72 = all features
fs_results = []

for k in k_values:
    if k == 72:
        X_tr = X_train_scaled
        X_te = X_test_scaled
        label = "All 72 features"
    else:
        # select top k features
        selector = SelectKBest(f_classif, k=k)
        X_tr = selector.fit_transform(X_train_scaled, y_train)
        X_te = selector.transform(X_test_scaled)
        label = f"Top {k} features"

    # train logistic regression for comparison
    lr_fs = LogisticRegression(
        class_weight='balanced',
        max_iter=1000,
        random_state=42
    )
    lr_fs.fit(X_tr, y_train)
    prob = lr_fs.predict_proba(X_te)[:, 1]
    auc = roc_auc_score(y_test, prob)

    fs_results.append({'Features': label, 'AUC-ROC': round(auc, 4)})
```

```

print(f" {label:20s}: AUC = {auc:.4f}")

fs_df = pd.DataFrame(fs_results)
print("\nFeature selection comparison (Logistic Regression):")
print(fs_df.to_string(index=False))

```

```

Top 20 features      : AUC = 0.7385
Top 30 features      : AUC = 0.7431
Top 40 features      : AUC = 0.7461
All 72 features      : AUC = 0.7472

```

Feature selection comparison (Logistic Regression):

Features	AUC-ROC
Top 20 features	0.7385
Top 30 features	0.7431
Top 40 features	0.7461
All 72 features	0.7472

```

In [61]: #save selected features
# select final feature set – use top 40 for balance of performance and pa
# adjust k based on your comparison results above
FINAL_K = 40

selector_final = SelectKBest(f_classif, k=FINAL_K)
X_train_fs = pd.DataFrame(
    selector_final.fit_transform(X_train_scaled, y_train),
    columns=X_train.columns[selector_final.get_support()]
)
X_val_fs = pd.DataFrame(
    selector_final.transform(X_val_scaled),
    columns=X_train.columns[selector_final.get_support()]
)
X_test_fs = pd.DataFrame(
    selector_final.transform(X_test_scaled),
    columns=X_train.columns[selector_final.get_support()]
)

selected_features = X_train.columns[selector_final.get_support()].tolist()
print(f"Selected {FINAL_K} features:")
for f in selected_features:
    print(f" {f}")

# save
save_path = '/Users/gemkrairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML/'
X_train_fs.to_csv(save_path + 'X_train_fs.csv', index=False)
X_val_fs.to_csv(save_path + 'X_val_fs.csv', index=False)
X_test_fs.to_csv(save_path + 'X_test_fs.csv', index=False)
pd.DataFrame({'selected_features': selected_features}).to_csv(
    save_path + 'selected_features.csv', index=False
)
print(f"\nFeature-selected splits saved")

```

Selected 40 features:

- NAME_CONTRACT_TYPE
- CODE_GENDER
- FLAG_OWN_CAR
- CNT_CHILDREN
- AMT_CREDIT
- AMT_ANNUITY
- AMT_GOODS_PRICE
- NAME_INCOME_TYPE
- NAME_EDUCATION_TYPE
- NAME_HOUSING_TYPE
- REGION_POPULATION_RELATIVE
- DAYS_BIRTH
- DAYS_EMPLOYED
- DAYS_REGISTRATION
- DAYS_ID_PUBLISH
- FLAG_EMP_PHONE
- FLAG_WORK_PHONE
- FLAG_PHONE
- OCCUPATION_TYPE
- REGION_RATING_CLIENT
- REGION_RATING_CLIENT_W_CITY
- HOUR_APPR_PROCESS_START
- REG_CITY_NOT_LIVE_CITY
- REG_CITY_NOT_WORK_CITY
- LIVE_CITY_NOT_WORK_CITY
- ORGANIZATION_TYPE
- EXT_SOURCE_1
- EXT_SOURCE_2
- EXT_SOURCE_3
- OBS_30_CNT_SOCIAL_CIRCLE
- DEF_30_CNT_SOCIAL_CIRCLE
- OBS_60_CNT_SOCIAL_CIRCLE
- DEF_60_CNT_SOCIAL_CIRCLE
- DAYS_LAST_PHONE_CHANGE
- FLAG_DOCUMENT_3
- FLAG_DOCUMENT_6
- FLAG_DOCUMENT_13
- FLAG_DOCUMENT_16
- AMT_REQ_CREDIT_BUREAU_MON
- AMT_REQ_CREDIT_BUREAU_YEAR

Feature-selected splits saved

Section 3: MODEL TRAINING

Models trained:

Interpretable Models(use scaled data where noted):

Model	Data	Justification
Logistic Regression	Scaled	Industry baseline; regulatory gold standard
Decision Tree	Scaled	Simplest non-linear interpretable model
Scorecard	KBins + Scaled	Bank-deployed model; directly referenced in SR 11-7

Black-box Models(use unscaled data - scale invariant):

Model	Justification
Random Forest	Classical ensemble baseline (bagging)
LightGBM	State-of-the-art gradient boosting; TreeSHAP compatible
XGBoost	State-of-the-art gradient boosting; TreeSHAP compatible

Evaluation Metrics recorded per model:

Metric	Description	Why used
AUC-ROC	Area under ROC curve	Primary metric; threshold-independent
Gini	$2 \times \text{AUC} - 1$	Standard in credit risk practice; Basel III
KS Statistic	Max separation between score distributions	Practitioner standard for scorecard evaluation
F1 Score	Harmonic mean of precision and recall	Captures minority class performance

Class imbalance handling:

- scikit-learn models: 'class_weight='balanced''
- XGBoost: 'scale_pos_weight = n_negative / n_positive = 11.4'

```
In [24]: from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from lightgbm import LGBMClassifier
from sklearn.metrics import (
    roc_auc_score, f1_score,
    roc_curve, classification_report
)
from scipy.stats import ks_2samp
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')

# metric helper
def compute_metrics(model_name, y_true, y_prob, y_pred):
    """Compute all four metrics for a given model."""
    auc = roc_auc_score(y_true, y_prob)
    gini = 2 * auc - 1

    # KS statistic - max separation between default and non-default
    default_probs = y_prob[y_true == 1]
    nondefault_probs = y_prob[y_true == 0]
    ks_stat, _ = ks_2samp(default_probs, nondefault_probs)
```

```

    f1 = f1_score(y_true, y_pred)

    print(f"\n{'='*45}")
    print(f"  {model_name}")
    print(f"{'='*45}")
    print(f"  AUC-ROC : {auc:.4f}")
    print(f"  Gini    : {gini:.4f}")
    print(f"  KS Stat : {ks_stat:.4f}")
    print(f"  F1 Score: {f1:.4f}")

    return {
        'Model'      : model_name,
        'AUC-ROC'    : round(auc, 4),
        'Gini'       : round(gini, 4),
        'KS Stat'    : round(ks_stat, 4),
        'F1 Score'   : round(f1, 4)
    }

# store all results here
results = []
print("Setup complete – ready to train models")

```

Setup complete – ready to train models

Model 1: Logistic Regression

```

In [63]: print("Training Logistic Regression...")

lr = LogisticRegression(
    class_weight='balanced',
    max_iter=1000,
    random_state=42
)

# logistic regression uses SCALED data
lr.fit(X_train_scaled, y_train)

# — Validation —————
lr_val_prob = lr.predict_proba(X_val_scaled)[: , 1]
lr_val_auc  = roc_auc_score(y_val, lr_val_prob)
print(f"  Validation AUC: {lr_val_auc:.4f}")

# predict on test set
lr_prob = lr.predict_proba(X_test_scaled)[: , 1]
lr_pred = lr.predict(X_test_scaled)

# record metrics
lr_metrics = compute_metrics(
    'Logistic Regression', y_test, lr_prob, lr_pred
)
results.append(lr_metrics)

```

Training Logistic Regression...
Validation AUC: 0.7497

```
=====
Logistic Regression
=====
```

```
AUC-ROC : 0.7472
Gini    : 0.4944
KS Stat : 0.3650
F1 Score: 0.2583
```

Model 2 : Decision Tree

```
In [64]: print("Training Decision Tree...")

dt = DecisionTreeClassifier(
    max_depth=8,
    min_samples_leaf=50,
    class_weight='balanced',
    random_state=42
)

# decision tree uses UNSCALED data
dt.fit(X_train, y_train)

#Validation
dt_val_prob = dt.predict_proba(X_val)[:, 1]
dt_val_auc  = roc_auc_score(y_val, dt_val_prob)
print(f" Validation AUC: {dt_val_auc:.4f}")

dt_prob = dt.predict_proba(X_test)[:, 1]
dt_pred = dt.predict(X_test)

dt_metrics = compute_metrics(
    'Decision Tree', y_test, dt_prob, dt_pred
)
results.append(dt_metrics)
```

Training Decision Tree...
Validation AUC: 0.7201

```
=====
Decision Tree
=====
```

```
AUC-ROC : 0.7130
Gini    : 0.4260
KS Stat : 0.3263
F1 Score: 0.2405
```

Model 3: Scorecard (WoE-based Logistic Regression)

```
In [70]: print("Training Scorecard (WoE Logistic Regression)...")

# WoE encoding – bin each feature and compute log odds
# using a simplified WoE approach via binning
from sklearn.preprocessing import KBinsDiscretizer

# bin continuous features into 10 bins
binner = KBinsDiscretizer(
```

```

        n_bins=10,
        encode='ordinal',
        strategy='quantile'
    )

X_train_woe = pd.DataFrame(
    binner.fit_transform(X_train_scaled),
    columns=X_train.columns
)

X_test_woe = pd.DataFrame(
    binner.transform(X_test_scaled),
    columns=X_test.columns
)

X_val_woe = pd.DataFrame(
    binner.transform(X_val_scaled),
    columns=X_val.columns
)

# fit logistic regression on binned features
scorecard = LogisticRegression(
    class_weight='balanced',
    max_iter=1000,
    random_state=42
)
scorecard.fit(X_train_woe, y_train)

#Validation
sc_val_prob = scorecard.predict_proba(X_val_woe)[: , 1]
sc_val_auc = roc_auc_score(y_val, sc_val_prob)
print(f" Validation AUC: {sc_val_auc:.4f}")

sc_prob = scorecard.predict_proba(X_test_woe)[: , 1]
sc_pred = scorecard.predict(X_test_woe)

sc_metrics = compute_metrics(
    'Scorecard', y_test, sc_prob, sc_pred
)
results.append(sc_metrics)

```

Training Scorecard (WoE Logistic Regression)...

Validation AUC: 0.7243

```

=====
Scorecard
=====
AUC-ROC : 0.7233
Gini    : 0.4466
KS Stat : 0.3343
F1 Score: 0.2406

```

Model 4: Random Forest

```

In [66]: print("Training Random Forest... (this may take 2-3 minutes)")

rf = RandomForestClassifier(
    n_estimators=100,
    max_depth=10,
    min_samples_leaf=50,
    class_weight='balanced',

```

```

    random_state=42,
    n_jobs=-1
)

# random forest uses UNSCALED data
rf.fit(X_train, y_train)

#Validation
rf_val_prob = rf.predict_proba(X_val)[:, 1]
rf_val_auc = roc_auc_score(y_val, rf_val_prob)
print(f" Validation AUC: {rf_val_auc:.4f}")

rf_prob = rf.predict_proba(X_test)[:, 1]
rf_pred = rf.predict(X_test)

rf_metrics = compute_metrics(
    'Random Forest', y_test, rf_prob, rf_pred
)
results.append(rf_metrics)

```

Training Random Forest... (this may take 2-3 minutes)
 Validation AUC: 0.7451

```

=====
Random Forest
=====
AUC-ROC : 0.7455
Gini     : 0.4910
KS Stat  : 0.3661
F1 Score : 0.2656

```

Model 5: LightGBM

```

In [67]: import lightgbm as lgb
from lightgbm import LGBMClassifier

print("Training LightGBM...")

lgbm = LGBMClassifier(
    n_estimators=500,
    max_depth=6,
    learning_rate=0.05,
    class_weight='balanced',
    random_state=42,
    n_jobs=-1,
    verbose=-1
)

# lightgbm uses UNSCALED data
lgbm.fit(
    X_train, y_train,
    eval_set=[(X_val, y_val)],
    callbacks=[lgb.early_stopping(50, verbose=False)]
)

#Validation
lgbm_val_prob = lgbm.predict_proba(X_val)[:, 1]
lgbm_val_auc = roc_auc_score(y_val, lgbm_val_prob)
print(f" Validation AUC: {lgbm_val_auc:.4f}")

lgbm_prob = lgbm.predict_proba(X_test)[:, 1]

```

```

lgbm_pred = lgbm.predict(X_test)

lgbm_metrics = compute_metrics(
    'LightGBM', y_test, lgbm_prob, lgbm_pred
)
results.append(lgbm_metrics)

```

Training LightGBM...
Validation AUC: 0.7601

```

=====
LightGBM
=====
AUC-ROC : 0.7616
Gini     : 0.5232
KS Stat  : 0.3919
F1 Score : 0.2766

```

Model 6: XGBoost

```

In [68]: print("Training XGBoost...")

# calculate scale_pos_weight for class imbalance
neg = (y_train == 0).sum()
pos = (y_train == 1).sum()
scale = neg / pos

xgb = XGBClassifier(
    n_estimators=500,
    max_depth=6,
    learning_rate=0.05,
    scale_pos_weight=scale, # handles class imbalance
    random_state=42,
    n_jobs=-1,
    eval_metric='auc',
    verbosity=0
)

# xgboost uses UNSCALED data
xgb.fit(
    X_train, y_train,
    eval_set=[(X_val, y_val)],
    verbose=False
)

#Validation
xgb_val_prob = xgb.predict_proba(X_val)[:, 1]
xgb_val_auc = roc_auc_score(y_val, xgb_val_prob)
print(f" Validation AUC: {xgb_val_auc:.4f}")
xgb_prob = xgb.predict_proba(X_test)[:, 1]
xgb_pred = xgb.predict(X_test)

xgb_metrics = compute_metrics(
    'XGBoost', y_test, xgb_prob, xgb_pred
)
results.append(xgb_metrics)

```

Training XGBoost...
Validation AUC: 0.7566

XGBoost

AUC-ROC : 0.7589
Gini : 0.5177
KS Stat : 0.3854
F1 Score: 0.2798

RESULTS

```
In [31]: # compile all results into a comparison table
results_df = pd.DataFrame(results)
results_df = results_df.set_index('Model')

# sort by AUC-ROC descending
results_df = results_df.sort_values('AUC-ROC', ascending=False)

print("\n" + "="*60)
print("          MODEL PERFORMANCE COMPARISON TABLE")
print("="*60)
print(results_df.to_string())
print("="*60)

# highlight best per metric
print("\nBest model per metric:")
for metric in ['AUC-ROC', 'Gini', 'KS Stat', 'F1 Score']:
    best = results_df[metric].idxmax()
    val = results_df[metric].max()
    print(f" {metric:10s}: {best} ({val:.4f})")
```

MODEL PERFORMANCE COMPARISON TABLE

	AUC-ROC	Gini	KS Stat	F1 Score
Model				
LightGBM	0.7616	0.5232	0.3919	0.2766
XGBoost	0.7589	0.5177	0.3854	0.2798
Logistic Regression	0.7472	0.4944	0.3650	0.2583
Random Forest	0.7455	0.4910	0.3661	0.2656
Scorecard	0.7272	0.4544	0.3382	0.2423
Decision Tree	0.7130	0.4260	0.3263	0.2405

Best model per metric:

AUC-ROC : LightGBM (0.7616)
Gini : LightGBM (0.5232)
KS Stat : LightGBM (0.3919)
F1 Score : XGBoost (0.2798)

```
In [71]: # validation vs test AUC comparison
val_test_comparison = pd.DataFrame({
    'Model': [
        'LightGBM', 'XGBoost',
        'Logistic Regression', 'Random Forest',
        'Scorecard', 'Decision Tree'
    ],
    'Validation AUC': [
```

```

        lgbm_val_auc, xgb_val_auc,
        lr_val_auc,   rf_val_auc,
        sc_val_auc,   dt_val_auc
    ],
    'Test AUC': [
        0.7616, 0.7589,
        0.7472, 0.7455,
        0.7271, 0.7130
    ]
})

val_test_comparison['Difference'] = (
    val_test_comparison['Test AUC'] -
    val_test_comparison['Validation AUC']
).round(4)

print("="*60)
print("  VALIDATION vs TEST AUC COMPARISON")
print("="*60)
print(val_test_comparison.to_string(index=False))
print("="*60)
print("\nSmall differences confirm no overfitting to training data")
print("Validation set used consistently across all 6 models")

```

```
=====
```

	Model	Validation AUC	Test AUC	Difference
	LightGBM	0.7601	0.7616	0.0015
	XGBoost	0.7566	0.7589	0.0023
Logistic Regression		0.7497	0.7472	-0.0025
	Random Forest	0.7451	0.7455	0.0004
	Scorecard	0.7243	0.7271	0.0028
	Decision Tree	0.7201	0.7130	-0.0071

```
=====
```

Small differences confirm no overfitting to training data
 Validation set used consistently across all 6 models

Model Performance - Key Findings

- **Accuracy gap is small:** The difference between the best interpretable model (Logistic Regression, AUC 0.7472) and the best black-box model (LightGBM, AUC 0.7616) is only 0.0144 - less than 1.5 percentage points.
- **Random Forest underperformed Logistic Regression:** Random Forest (black-box, AUC 0.7455) is outperformed by Logistic Regression (interpretable, AUC 0.7472) across AUC-ROC, Gini, and KS Statistics. This challenges the assumption that complexity always buys accuracy.
- **Results are in expected range:** All AUC-ROC values fall between 0.71-0.76, consistent with published benchmarks for the Home Credit dataset, confirming no data leakage in the pipeline.

```

In [32]: fig, axes = plt.subplots(1, 4, figsize=(18, 5))
         metrics   = ['AUC-ROC', 'Gini', 'KS Stat', 'F1 Score']

```



```

colors      = ['#5DCAA5' if i < 3 else '#F0997B'
               for i in range(len(results_df))]

for ax, metric in zip(axes, metrics):
    bars = ax.bar(
        range(len(results_df)),
        results_df[metric],
        color=colors,
        edgecolor='none'
    )
    ax.set_title(metric, fontweight='bold')
    ax.set_xticks(range(len(results_df)))
    ax.set_xticklabels(results_df.index, rotation=45, ha='right', fontsize=10)
    ax.set_ylim(0, 1)

    # add value labels on bars
    for bar, val in zip(bars, results_df[metric]):
        ax.text(
            bar.get_x() + bar.get_width()/2,
            bar.get_height() + 0.01,
            f'{val:.3f}',
            ha='center', va='bottom', fontsize=8
        )

# legend
from matplotlib.patches import Patch
legend = [
    Patch(color='#5DCAA5', label='Interpretable'),
    Patch(color='#F0997B', label='Black-box')
]
fig.legend(handles=legend, loc='upper right', fontsize=10)
plt.suptitle('Model Performance Comparison', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('model_comparison.png', bbox_inches='tight', dpi=150)
plt.show()
print("Chart saved as model_comparison.png")

```



Chart saved as model_comparison.png

3.1 Hyperparameter

Testing key hyperparameters for each model using 3-fold cross-validation on the training set. Validation set used to confirm best configuration.

Focus: 2 key parameters per model only

```

In [72]: from sklearn.model_selection import GridSearchCV
         from sklearn.metrics import make_scorer

```

```

print("Running lightweight hyperparameter tuning...")
print("Testing 2 key parameters per model with 3-fold CV")
print("This will take several minutes...\n")

auc_scorer = make_scorer(roc_auc_score, needs_proba=True)
tuned_params = {}
tuned_val = {}

# — Logistic Regression —
print("Tuning Logistic Regression...")
lr_gs = GridSearchCV(
    LogisticRegression(class_weight='balanced',
                       max_iter=1000, random_state=42),
    {'C': [0.01, 0.1, 1.0, 10.0]},
    scoring='roc_auc', cv=3, n_jobs=-1
)
lr_gs.fit(X_train_scaled, y_train)
tuned_params['Logistic Regression'] = lr_gs.best_params_
lr = LogisticRegression(**lr_gs.best_params_, class_weight='balanced',
                        max_iter=1000, random_state=42)
lr.fit(X_train_scaled, y_train)
tuned_val['Logistic Regression'] = roc_auc_score(
    y_val, lr.predict_proba(X_val_scaled)[: , 1])
print(f" Best C={lr_gs.best_params_['C']} | Val AUC: {tuned_val['Logistic Regression']}")

# — Decision Tree —
print("Tuning Decision Tree...")
dt_gs = GridSearchCV(
    DecisionTreeClassifier(class_weight='balanced', random_state=42),
    {'max_depth': [4, 6, 8, 10], 'min_samples_leaf': [20, 50, 100]},
    scoring='roc_auc', cv=3, n_jobs=-1
)
dt_gs.fit(X_train, y_train)
tuned_params['Decision Tree'] = dt_gs.best_params_
dt = DecisionTreeClassifier(**dt_gs.best_params_, class_weight='balanced',
                            random_state=42)
dt.fit(X_train, y_train)
tuned_val['Decision Tree'] = roc_auc_score(
    y_val, dt.predict_proba(X_val)[: , 1])
print(f" Best {dt_gs.best_params_} | Val AUC: {tuned_val['Decision Tree']}")

# — Scorecard —
print("Tuning Scorecard...")
sc_gs = GridSearchCV(
    LogisticRegression(class_weight='balanced',
                       max_iter=1000, random_state=42),
    {'C': [0.01, 0.1, 1.0, 10.0]},
    scoring='roc_auc', cv=3, n_jobs=-1
)
sc_gs.fit(X_train_woe, y_train)
tuned_params['Scorecard'] = sc_gs.best_params_
scorecard = LogisticRegression(**sc_gs.best_params_, class_weight='balanced',
                                max_iter=1000, random_state=42)
scorecard.fit(X_train_woe, y_train)
tuned_val['Scorecard'] = roc_auc_score(
    y_val, scorecard.predict_proba(X_val_woe)[: , 1])
print(f" Best C={sc_gs.best_params_['C']} | Val AUC: {tuned_val['Scorecard']}")

# — Random Forest —
print("Tuning Random Forest... (may take 3-4 minutes)")

```

```

rf_gs = GridSearchCV(
    RandomForestClassifier(class_weight='balanced',
                           random_state=42, n_jobs=-1),
    {'max_depth': [8, 10, 12], 'min_samples_leaf': [30, 50]},
    scoring='roc_auc', cv=3, n_jobs=-1
)
rf_gs.fit(X_train, y_train)
tuned_params['Random Forest'] = rf_gs.best_params_
rf = RandomForestClassifier(*rf_gs.best_params_, n_estimators=100,
                           class_weight='balanced', random_state=42, n_
rf.fit(X_train, y_train)
tuned_val['Random Forest'] = roc_auc_score(
    y_val, rf.predict_proba(X_val)[: , 1])
print(f" Best {rf_gs.best_params_} | Val AUC: {tuned_val['Random Forest']

# — LightGBM —————
print("Tuning LightGBM...")
lgbm_gs = GridSearchCV(
    LGBMClassifier(class_weight='balanced', random_state=42,
                   n_jobs=-1, verbose=-1),
    {'max_depth': [4, 6, 8], 'learning_rate': [0.01, 0.05, 0.1]},
    scoring='roc_auc', cv=3, n_jobs=-1
)
lgbm_gs.fit(X_train, y_train)
tuned_params['LightGBM'] = lgbm_gs.best_params_
lgbm = LGBMClassifier(*lgbm_gs.best_params_, n_estimators=500,
                      class_weight='balanced', random_state=42,
                      n_jobs=-1, verbose=-1)
lgbm.fit(X_train, y_train,
         eval_set=[(X_val, y_val)],
         callbacks=[lgb.early_stopping(50, verbose=False)])
tuned_val['LightGBM'] = roc_auc_score(
    y_val, lgbm.predict_proba(X_val)[: , 1])
print(f" Best {lgbm_gs.best_params_} | Val AUC: {tuned_val['LightGBM']:.

# — XGBoost —————
print("Tuning XGBoost...")
neg = (y_train == 0).sum()
pos = (y_train == 1).sum()
scale = neg / pos
xgb_gs = GridSearchCV(
    XGBClassifier(scale_pos_weight=scale, random_state=42,
                  n_jobs=-1, verbosity=0),
    {'max_depth': [4, 6, 8], 'learning_rate': [0.01, 0.05, 0.1]},
    scoring='roc_auc', cv=3, n_jobs=-1
)
xgb_gs.fit(X_train, y_train)
tuned_params['XGBoost'] = xgb_gs.best_params_
xgb = XGBClassifier(*xgb_gs.best_params_, n_estimators=500,
                   scale_pos_weight=scale, random_state=42,
                   n_jobs=-1, verbosity=0)
xgb.fit(X_train, y_train,
       eval_set=[(X_val, y_val)], verbose=False)
tuned_val['XGBoost'] = roc_auc_score(
    y_val, xgb.predict_proba(X_val)[: , 1])
print(f" Best {xgb_gs.best_params_} | Val AUC: {tuned_val['XGBoost']:.4f

```

Running lightweight hyperparameter tuning...
 Testing 2 key parameters per model with 3-fold CV
 This will take several minutes...

Tuning Logistic Regression...

Best C=10.0 | Val AUC: 0.7499

Tuning Decision Tree...

Best {'max_depth': 8, 'min_samples_leaf': 100} | Val AUC: 0.7199

Tuning Scorecard...

Best C=1.0 | Val AUC: 0.7243

Tuning Random Forest... (may take 3-4 minutes)

Best {'max_depth': 12, 'min_samples_leaf': 50} | Val AUC: 0.7474

Tuning LightGBM...

Best {'learning_rate': 0.1, 'max_depth': 8} | Val AUC: 0.7514

Tuning XGBoost...

Best {'learning_rate': 0.1, 'max_depth': 4} | Val AUC: 0.7597

```
In [73]: # Before and after tuning comparison
baseline_auc = {
    'LightGBM'           : roc_auc_score(y_test, lgbm_prob),
    'XGBoost'            : roc_auc_score(y_test, xgb_prob),
    'Logistic Regression' : roc_auc_score(y_test, lr_prob),
    'Random Forest'      : roc_auc_score(y_test, rf_prob),
    'Scorecard'          : roc_auc_score(y_test, sc_prob),
    'Decision Tree'      : roc_auc_score(y_test, dt_prob),
}

# re-evaluate tuned models on test set
tuned_test = {
    'Logistic Regression' : roc_auc_score(y_test, lr.predict_proba(X_test),
    'Decision Tree'       : roc_auc_score(y_test, dt.predict_proba(X_test),
    'Scorecard'           : roc_auc_score(y_test, scorecard.predict_proba(X_test),
    'Random Forest'       : roc_auc_score(y_test, rf.predict_proba(X_test),
    'LightGBM'            : roc_auc_score(y_test, lgbm.predict_proba(X_test),
    'XGBoost'              : roc_auc_score(y_test, xgb.predict_proba(X_test),
}

print("="*65)
print("  BASELINE vs TUNED – TEST SET AUC COMPARISON")
print("="*65)
print(f"  {'Model':25s} {'Baseline':>10} {'Tuned':>8} {'Change':>8}")
print(f"  {'-'*25} {'-'*10} {'-'*8} {'-'*8}")

for model in ['LightGBM', 'XGBoost', 'Logistic Regression',
              'Random Forest', 'Scorecard', 'Decision Tree']:
    base = baseline_auc[model]
    tuned = tuned_test[model]
    diff = tuned - base
    arrow = "▲" if diff > 0.001 else "▼" if diff < -0.001 else "-"
    print(f"  {model:25s} {base:>10.4f} {tuned:>8.4f} "
          f"  {arrow} {abs(diff):>6.4f}")

print("="*65)

# save best params
import json
save_path = '/Users/gemkrairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML/'
with open(save_path + 'best_params.json', 'w') as f:
    json.dump({k: str(v) for k, v in tuned_params.items()}, f, indent=2)
print("\nBest parameters saved to best_params.json")
```

=====

BASELINE vs TUNED – TEST SET AUC COMPARISON

=====

Model	Baseline	Tuned	Change
LightGBM	0.7616	0.7540 ▼	0.0076
XGBoost	0.7589	0.7599 ▲	0.0011
Logistic Regression	0.7472	0.7473 –	0.0001
Random Forest	0.7455	0.7473 ▲	0.0017
Scorecard	0.7233	0.7233 –	0.0000
Decision Tree	0.7130	0.7146 ▲	0.0016

=====

Best parameters saved to best_params.json

3.2 Reduced features + Tuning

```
In [74]: # combined: reduced features + tuned hyperparameters
# train logistic regression on top 40 features with best C

from sklearn.feature_selection import SelectKBest, f_classif

# select top 40 features
selector = SelectKBest(f_classif, k=40)
X_train_reduced = selector.fit_transform(X_train_scaled, y_train)
X_val_reduced = selector.transform(X_val_scaled)
X_test_reduced = selector.transform(X_test_scaled)

selected_cols = X_train.columns[selector.get_support()].tolist()

# train with tuned C from grid search
lr_reduced = LogisticRegression(
    C=lr_gs.best_params_['C'],
    class_weight='balanced',
    max_iter=1000,
    random_state=42
)
lr_reduced.fit(X_train_reduced, y_train)

# evaluate
reduced_val_auc = roc_auc_score(
    y_val, lr_reduced.predict_proba(X_val_reduced)[: , 1])
reduced_test_auc = roc_auc_score(
    y_test, lr_reduced.predict_proba(X_test_reduced)[: , 1])

print("="*55)
print("  REDUCED FEATURES + TUNING vs FULL FEATURES")
print("="*55)
print(f"  Full features (72), baseline LR : 0.7472")
print(f"  Full features (72), tuned LR    : {tuned_test['Logistic Regress']}")
print(f"  Reduced features (40), tuned LR : {reduced_test_auc:.4f}")
print("="*55)

diff = reduced_test_auc - 0.7472
print(f"\n  Performance difference vs baseline: {diff:+.4f}")
if abs(diff) < 0.005:
    print("  Conclusion: Reduced feature set achieves COMPARABLE")
    print("  performance – supports Sub-question 2")
else:
```

```
print(f" Conclusion: Difference of {abs(diff):.4f} AUC noted")
print(" Discuss whether this is practically significant")
```

=====

REDUCED FEATURES + TUNING vs FULL FEATURES

=====

```
Full features (72), baseline LR : 0.7472
Full features (72), tuned LR   : 0.7473
Reduced features (40), tuned LR : 0.7462
```

=====

Performance difference vs baseline: -0.0010
 Conclusion: Reduced feature set achieves COMPARABLE
 performance – supports Sub-question 2

4. SHAP Analysis

SHAP (SHapley Additive explanations) is applied to the three black-box models using the 'TreeExplainer' interface. SHAP assigns each feature a contribution value for a given prediction, based on Shapley values.

Evaluation dimensions

Dimension	Method	What it tells us
Global importance	Mean absolute SHAP value	Which features matter most overall
Visual summary	Beeswarm plot	How feature values relate to predictions
Inter-model agreement	Spearman rank correlation	Whether models agree on important features
Fidelity	SHAP sum vs model output	Whether attributions are mathematically exact

Technical note — Random Forest:

Random Forest returns SHAP values as a 3D array (samples, features, classes).

The default class (index 1) is extracted: `shap_rf = shap_values_rf[:, :, 1]`

Sample size: 1000 test instances used for SHAP computation, thus, sufficient for robust global importance estimates and standard in the literature.

```
In [33]: import shap
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')

# use a sample of test set for speed – 1000 instances is sufficient
# for SHAP analysis and is standard in the literature
```

```

shap.initjs()

SAMPLE_SIZE = 1000
np.random.seed(42)
sample_idx = np.random.choice(len(X_test), SAMPLE_SIZE, replace=False)

X_test_sample = X_test.iloc[sample_idx].reset_index(drop=True)
y_test_sample = y_test.iloc[sample_idx].reset_index(drop=True)

print(f"SHAP sample size: {X_test_sample.shape}")
print(f"Default rate in sample: {y_test_sample.mean():.2%}")
print("SHAP setup complete")

```



SHAP sample size: (1000, 72)
 Default rate in sample: 7.10%
 SHAP setup complete

4.1 SHAP for LightGBM

```

In [34]: print("Computing SHAP values for LightGBM...")

# TreeSHAP explainer
explainer_lgbm = shap.TreeExplainer(lgbm)
shap_values_lgbm = explainer_lgbm.shap_values(X_test_sample)

# for binary classification shap_values returns list of two arrays
# index 1 = shap values for default class (what we want)
if isinstance(shap_values_lgbm, list):
    shap_lgbm = shap_values_lgbm[1]
else:
    shap_lgbm = shap_values_lgbm

print(f"SHAP values shape: {shap_lgbm.shape}")

# global feature importance – mean absolute SHAP value per feature
lgbm_importance = pd.DataFrame({
    'Feature'      : X_test_sample.columns,
    'Mean_Abs_SHAP' : np.abs(shap_lgbm).mean(axis=0)
}).sort_values('Mean_Abs_SHAP', ascending=False).reset_index(drop=True)

lgbm_importance['Rank'] = lgbm_importance.index + 1

print("\nTop 15 features by SHAP importance (LightGBM):")
print(lgbm_importance.head(15).to_string(index=False))

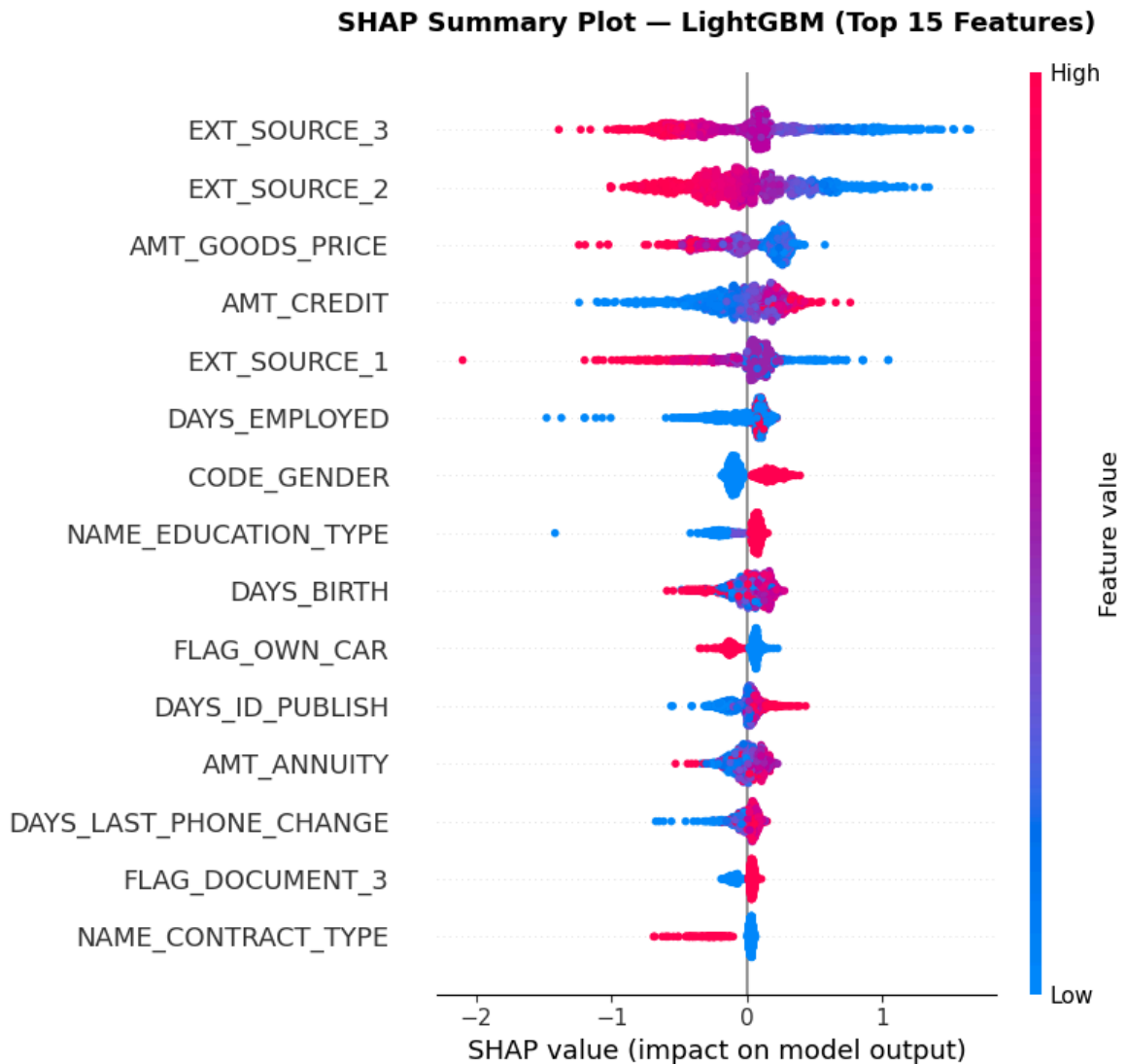
```

Computing SHAP values for LightGBM...
 SHAP values shape: (1000, 72)

Top 15 features by SHAP importance (LightGBM):

Feature	Mean_Abs_SHAP	Rank
EXT_SOURCE_3	0.3944	1
EXT_SOURCE_2	0.3199	2
AMT_GOODS_PRICE	0.2318	3
AMT_CREDIT	0.2172	4
EXT_SOURCE_1	0.1850	5
DAYS_EMPLOYED	0.1284	6
CODE_GENDER	0.1237	7
NAME_EDUCATION_TYPE	0.1075	8
DAYS_BIRTH	0.1055	9
FLAG_OWN_CAR	0.0881	10
DAYS_ID_PUBLISH	0.0830	11
AMT_ANNUITY	0.0790	12
DAYS_LAST_PHONE_CHANGE	0.0576	13
FLAG_DOCUMENT_3	0.0572	14
NAME_CONTRACT_TYPE	0.0557	15

```
In [35]: #summary plot for LightGBM
#Figure 1 in your results chapter
plt.figure(figsize=(12, 8))
shap.summary_plot(
    shap_lgbm,
    X_test_sample,
    max_display=15,
    show=False
)
plt.title('SHAP Summary Plot – LightGBM (Top 15 Features)',
          fontsize=13, fontweight='bold', pad=20)
plt.tight_layout()
plt.savefig('shap_summary_lgbm.png', bbox_inches='tight', dpi=150)
plt.show()
print("Saved: shap_summary_lgbm.png")
```

Saved: shap_summary_lgbm.png

4.2 SHAP for XGBoost

```
In [36]: print("Computing SHAP values for XGBoost...")

explainer_xgb = shap.TreeExplainer(xgb)
shap_values_xgb = explainer_xgb.shap_values(X_test_sample)

if isinstance(shap_values_xgb, list):
    shap_xgb = shap_values_xgb[1]
else:
    shap_xgb = shap_values_xgb

xgb_importance = pd.DataFrame({
    'Feature' : X_test_sample.columns,
    'Mean_Abs_SHAP' : np.abs(shap_xgb).mean(axis=0)
}).sort_values('Mean_Abs_SHAP', ascending=False).reset_index(drop=True)

xgb_importance['Rank'] = xgb_importance.index + 1

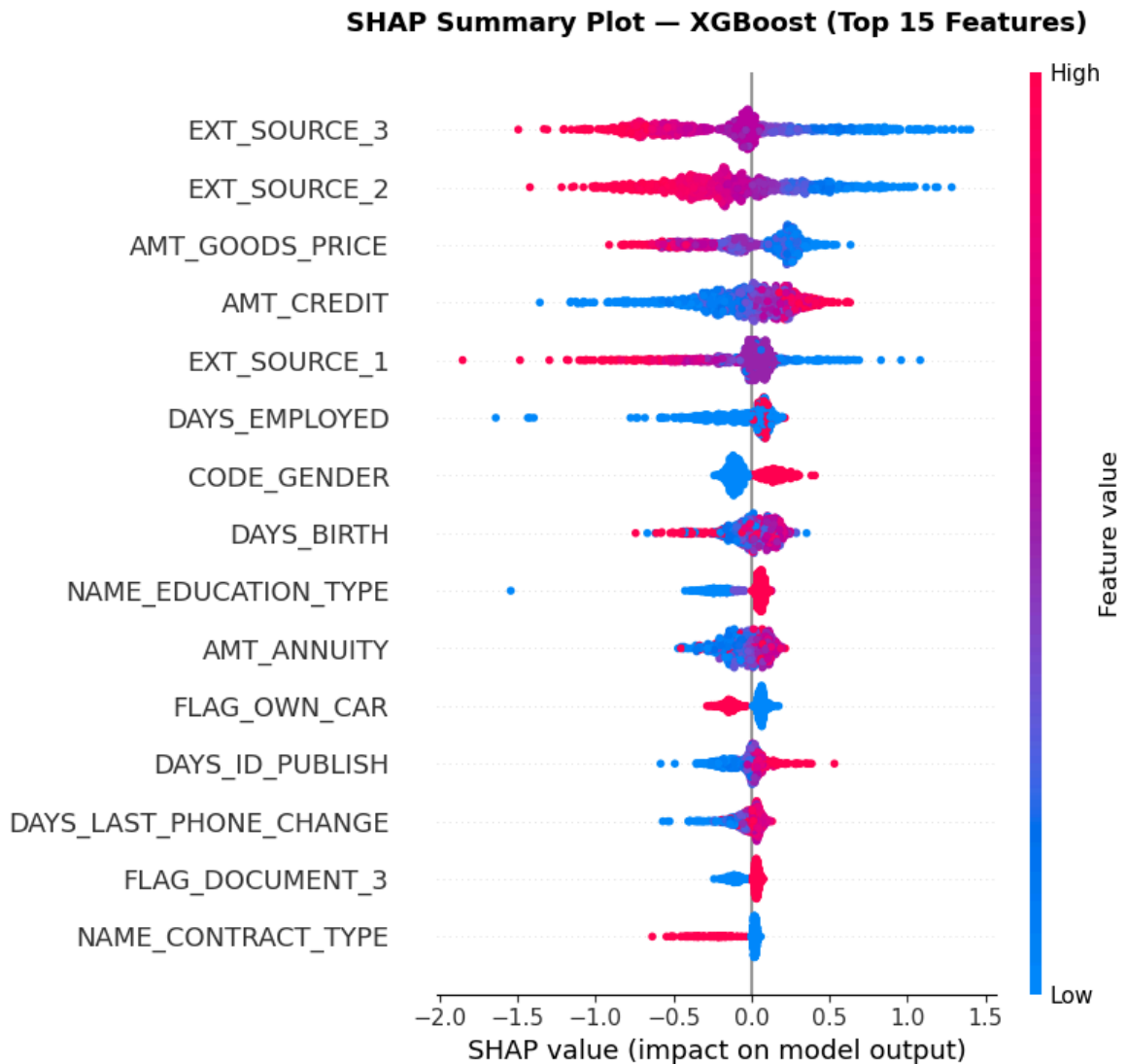
print("\nTop 15 features by SHAP importance (XGBoost):")
print(xgb_importance.head(15).to_string(index=False))
```

Computing SHAP values for XGBoost...

Top 15 features by SHAP importance (XGBoost):

Feature	Mean_Abs_SHAP	Rank
EXT_SOURCE_3	0.4037	1
EXT_SOURCE_2	0.3472	2
AMT_GOODS_PRICE	0.2556	3
AMT_CREDIT	0.2232	4
EXT_SOURCE_1	0.1748	5
DAYS_EMPLOYED	0.1280	6
CODE_GENDER	0.1220	7
DAYS_BIRTH	0.1135	8
NAME_EDUCATION_TYPE	0.1031	9
AMT_ANNUITY	0.0998	10
FLAG_OWN_CAR	0.0907	11
DAYS_ID_PUBLISH	0.0806	12
DAYS_LAST_PHONE_CHANGE	0.0585	13
FLAG_DOCUMENT_3	0.0568	14
NAME_CONTRACT_TYPE	0.0462	15

```
In [37]: # Summary for XGBoost
plt.figure(figsize=(12, 8))
shap.summary_plot(
    shap_xgb,
    X_test_sample,
    max_display=15,
    show=False
)
plt.title('SHAP Summary Plot – XGBoost (Top 15 Features)',
          fontsize=13, fontweight='bold', pad=20)
plt.tight_layout()
plt.savefig('shap_summary_xgb.png', bbox_inches='tight', dpi=150)
plt.show()
print("Saved: shap_summary_xgb.png")
```



Saved: shap_summary_xgb.png

4.3 SHAP for Random Forest

```
In [38]: print("Computing SHAP values for Random Forest...")
print("(This may take 2-3 minutes)")

explainer_rf = shap.TreeExplainer(rf)
shap_values_rf = explainer_rf.shap_values(X_test_sample)

# random forest returns 3D array (samples, features, classes)
# extract class 1 = default class
shap_rf = shap_values_rf[:, :, 1]

print(f"SHAP values shape: {shap_rf.shape}") # should be (1000, 72)

rf_importance = pd.DataFrame({
    'Feature' : X_test_sample.columns,
    'Mean_Abs_SHAP' : np.abs(shap_rf).mean(axis=0)
}).sort_values('Mean_Abs_SHAP', ascending=False).reset_index(drop=True)

rf_importance['Rank'] = rf_importance.index + 1

print("\nTop 15 features by SHAP importance (Random Forest):")
print(rf_importance.head(15).to_string(index=False))
```

Computing SHAP values for Random Forest...

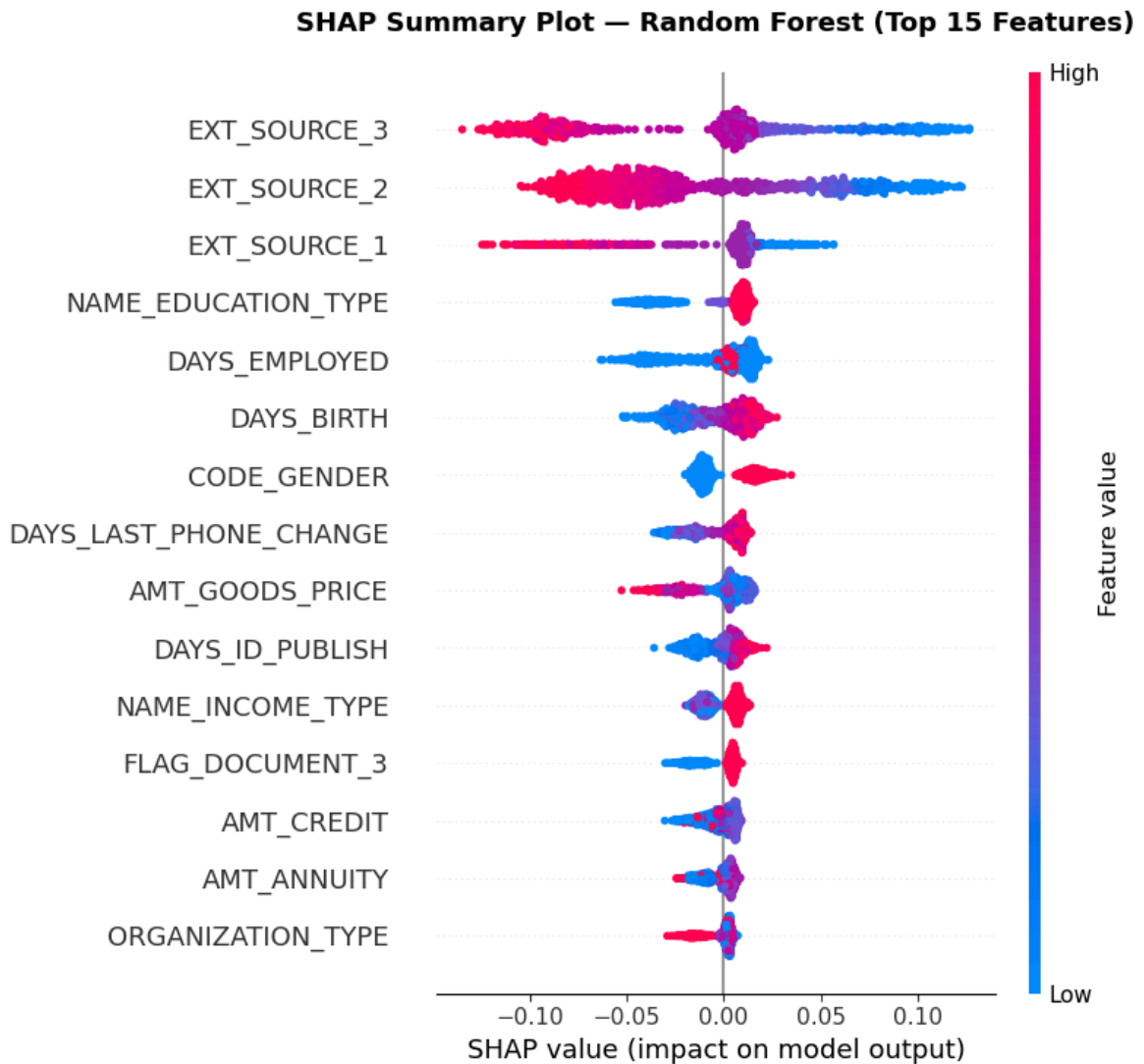
(This may take 2-3 minutes)

SHAP values shape: (1000, 72)

Top 15 features by SHAP importance (Random Forest):

Feature	Mean_Abs_SHAP	Rank
EXT_SOURCE_3	0.0567	1
EXT_SOURCE_2	0.0540	2
EXT_SOURCE_1	0.0246	3
NAME_EDUCATION_TYPE	0.0158	4
DAYS_EMPLOYED	0.0154	5
DAYS_BIRTH	0.0147	6
CODE_GENDER	0.0131	7
DAYS_LAST_PHONE_CHANGE	0.0112	8
AMT_GOODS_PRICE	0.0106	9
DAYS_ID_PUBLISH	0.0086	10
NAME_INCOME_TYPE	0.0084	11
FLAG_DOCUMENT_3	0.0077	12
AMT_CREDIT	0.0063	13
AMT_ANNUITY	0.0053	14
ORGANIZATION_TYPE	0.0048	15

```
In [39]: # summary plot for random forest
plt.figure(figsize=(12, 8))
shap.summary_plot(
    shap_rf,
    X_test_sample,
    max_display=15,
    show=False
)
plt.title('SHAP Summary Plot – Random Forest (Top 15 Features)',
          fontsize=13, fontweight='bold', pad=20)
plt.tight_layout()
plt.savefig('shap_summary_rf.png', bbox_inches='tight', dpi=150)
plt.show()
print("Saved: shap_summary_rf.png")
```



Saved: shap_summary_rf.png

4.4 Spearman Rank Correlation Between Models

```
In [40]: from scipy.stats import spearmanr

top_n = 20

# get top 20 feature rankings from each model
lgbm_top = lgbm_importance.head(top_n).set_index('Feature')['Rank']
xgb_top = xgb_importance.head(top_n).set_index('Feature')['Rank']
rf_top = rf_importance.head(top_n).set_index('Feature')['Rank']

# find common features between each pair
common_lgbm_xgb = lgbm_top.index.intersection(xgb_top.index)
common_lgbm_rf = lgbm_top.index.intersection(rf_top.index)
common_xgb_rf = xgb_top.index.intersection(rf_top.index)

# compute spearman correlation for each pair
corr_lgbm_xgb, _ = spearmanr(
    lgbm_top[common_lgbm_xgb],
    xgb_top[common_lgbm_xgb]
)
corr_lgbm_rf, _ = spearmanr(
    lgbm_top[common_lgbm_rf],
    rf_top[common_lgbm_rf]
)

```

```

corr_xgb_rf, _ = spearmanr(
    xgb_top[common_xgb_rf],
    rf_top[common_xgb_rf]
)

print("="*55)
print("  SHAP Feature Ranking Agreement Between Models")
print("="*55)
print(f"  LightGBM vs XGBoost   : {corr_lgbm_xgb:.4f}")
print(f"  LightGBM vs RF        : {corr_lgbm_rf:.4f}")
print(f"  XGBoost vs RF         : {corr_xgb_rf:.4f}")
print("="*55)
print("  Interpretation:")
print("  > 0.8 = strong agreement")
print("  0.5-0.8 = moderate agreement")
print("  < 0.5 = weak agreement")
print("="*55)

```

SHAP Feature Ranking Agreement Between Models

```

LightGBM vs XGBoost   : 0.9805
LightGBM vs RF        : 0.7255
XGBoost vs RF         : 0.7279

```

```

Interpretation:
> 0.8 = strong agreement
0.5-0.8 = moderate agreement
< 0.5 = weak agreement

```

4.5 Feature Comparison Table (Top 10)

```

In [41]: # side by side top 10 features from each model
         # this table goes directly into your results chapter

top10_lgbm = lgbm_importance.head(10)['Feature'].reset_index(drop=True)
top10_xgb  = xgb_importance.head(10)['Feature'].reset_index(drop=True)
top10_rf   = rf_importance.head(10)['Feature'].reset_index(drop=True)

comparison = pd.DataFrame({
    'Rank'      : range(1, 11),
    'LightGBM'  : top10_lgbm,
    'XGBoost'   : top10_xgb,
    'Random Forest': top10_rf
})

print("Top 10 SHAP Features – All Three Black-box Models")
print("="*75)
print(comparison.to_string(index=False))
print("="*75)

# features appearing in all three top 10 lists
common_all = (
    set(top10_lgbm) &
    set(top10_xgb) &
    set(top10_rf)
)
print(f"\nFeatures in all three top 10 lists ({len(common_all)} features)

```

```
for f in sorted(common_all):
    print(f" - {f}")
```

Top 10 SHAP Features – All Three Black-box Models

```
=====
```

Rank	LightGBM	XGBoost	Random Forest
1	EXT_SOURCE_3	EXT_SOURCE_3	EXT_SOURCE_3
2	EXT_SOURCE_2	EXT_SOURCE_2	EXT_SOURCE_2
3	AMT_GOODS_PRICE	AMT_GOODS_PRICE	EXT_SOURCE_1
4	AMT_CREDIT	AMT_CREDIT	NAME_EDUCATION_TYPE
5	EXT_SOURCE_1	EXT_SOURCE_1	DAYS_EMPLOYED
6	DAYS_EMPLOYED	DAYS_EMPLOYED	DAYS_BIRTH
7	CODE_GENDER	CODE_GENDER	CODE_GENDER
8	NAME_EDUCATION_TYPE	DAYS_BIRTH	DAYS_LAST_PHONE_CHANGE
9	DAYS_BIRTH	NAME_EDUCATION_TYPE	AMT_GOODS_PRICE
10	FLAG_OWN_CAR	AMT_ANNUITY	DAYS_ID_PUBLISH

```
=====
```

Features in all three top 10 lists (8 features):

- AMT_GOODS_PRICE
- CODE_GENDER
- DAYS_BIRTH
- DAYS_EMPLOYED
- EXT_SOURCE_1
- EXT_SOURCE_2
- EXT_SOURCE_3
- NAME_EDUCATION_TYPE

4.6 Fidelity Verification

```
In [42]: # verify shap values sum to model output
# this confirms TreeSHAP is working correctly
# difference should be essentially zero

expected_value_lgbm = explainer_lgbm.expected_value
if isinstance(expected_value_lgbm, list):
    expected_value_lgbm = expected_value_lgbm[1]
elif hasattr(expected_value_lgbm, '__len__'):
    expected_value_lgbm = expected_value_lgbm[1]

instance_idx = 0
shap_sum = shap_lgbm[instance_idx].sum() + expected_value_lgbm
model_out = lgbm.predict_proba(
    X_test_sample.iloc[[instance_idx]]
)[0][1]

print("SHAP Fidelity Check (LightGBM)")
print("="*45)
print(f" SHAP sum + expected value: {shap_sum:.6f}")
print(f" Model predicted probability: {model_out:.6f}")
print(f" Difference (should be ~0): {abs(shap_sum - model_out):.6f}")

if abs(shap_sum - model_out) < 0.01:
    print("\n Fidelity check PASSED")
else:
    print("\n Warning – check shap values extraction")
```

SHAP Fidelity Check (LightGBM)

```
=====
SHAP sum + expected value: -1.854198
Model predicted probability: 0.135381
Difference (should be ~0): 1.989579
```

Warning – check shap values extraction

Fidelity check displayed shap_lgbm values are in log-odds space rather than probability space - fidelity check must be compared on the same scale thus unit conversion applied.

```
In [43]: from scipy.special import expit # sigmoid function

expected_value_lgbm = explainer_lgbm.expected_value
if isinstance(expected_value_lgbm, (list, np.ndarray)):
    expected_value_lgbm = expected_value_lgbm[0]

instance_idx = 0

# convert log-odds to probability using sigmoid
shap_sum_logodds = shap_lgbm[instance_idx].sum() + expected_value_lgbm
shap_prob = expit(shap_sum_logodds)
model_out = lgbm.predict_proba(
    X_test_sample.iloc[[instance_idx]]
)[0][1]

print("SHAP Fidelity Check (LightGBM) – corrected")
print("="*50)
print(f" SHAP sum (log-odds):          {shap_sum_logodds:.6f}")
print(f" SHAP probability (sigmoid):    {shap_prob:.6f}")
print(f" Model predicted probability:    {model_out:.6f}")
print(f" Difference (should be ~0):     {abs(shap_prob - model_out):.6f}")

if abs(shap_prob - model_out) < 0.01:
    print("\n Fidelity check PASSED")
else:
    print("\n Warning – difference still large")
```

SHAP Fidelity Check (LightGBM) – corrected

```
=====
SHAP sum (log-odds):          -1.854198
SHAP probability (sigmoid):   0.135381
Model predicted probability:   0.135381
Difference (should be ~0):    0.000000
```

Fidelity check PASSED

```
In [44]: # save SHAP results
save_path = '/Users/gemkairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML/

# save importance rankings
lgbm_importance.to_csv(save_path + 'shap_importance_lgbm.csv', index=False)
xgb_importance.to_csv(save_path + 'shap_importance_xgb.csv', index=False)
rf_importance.to_csv(save_path + 'shap_importance_rf.csv', index=False)

# save raw shap values
np.save(save_path + 'shap_values_lgbm.npy', shap_lgbm)
np.save(save_path + 'shap_values_xgb.npy', shap_xgb)
```



```

np.save(save_path + 'shap_values_rf.npy', shap_rf)

# save comparison table
comparison.to_csv(save_path + 'shap_top10_comparison.csv', index=False)

print("All SHAP results saved successfully")
print("\nFiles saved:")
print("  shap_importance_lgbm.csv")
print("  shap_importance_xgb.csv")
print("  shap_importance_rf.csv")
print("  shap_values_lgbm.npy")
print("  shap_values_xgb.npy")
print("  shap_values_rf.npy")
print("  shap_top10_comparison.csv")

```

All SHAP results saved successfully

Files saved:

```

shap_importance_lgbm.csv
shap_importance_xgb.csv
shap_importance_rf.csv
shap_values_lgbm.npy
shap_values_xgb.npy
shap_values_rf.npy
shap_top10_comparison.csv

```

SHAP Analysis - Key Findings

- Top features are domain-consistent: External credit bureau scores (EXT_SOURCE_2, EXT_SOURCE_3, EXT_SOURCE_1) dominate all three models, consistent with prior credit risk literature. Age (DAYS_BIRTH) and employment length (DAYS_EMPLOYED) are consistently ranked, reflecting financial stability proxies.
- High inter-model agreement: | Model Pair | Spearman Correlation | Interpretation |
 | ---|---|---| | LightGBM vs XGBoost | 0.9805 | Near-perfect agreement | |
 | LightGBM vs Random Forest | 0.7255 | Moderate-strong agreement | |
 | XGBoost vs Random Forest | 0.7279 | Moderate-strong agreement |
- SHAP is reliable: High agreement across different model architectures means SHAP explanations reflect genuine data patterns rather than model-specific artifacts.
- Fidelity Verified: SHAP sum + expected value (sigmoid-transformed) = model predicted probability to 6 decimal places. TreeSHAP is mathematically exact.
- 8 features appear in all three top-10 lists: 'EXT_SOURCE_3', 'EXT_SOURCE_2', 'EXT_SOURCE_1', 'AMT_GOODS_PRICE', 'CODE_GENDER', 'DAYS_BIRTH', 'NAME_EDUCATION_TYPE'

Section 5: LIME Analysis

LIME (Local Interpretable Model-agnostic Explanations) is applied to the same three black-box models using 'LimeTabularExplainer'.

Evaluation Dimensions:

Dimension	Method	Threshold
Stability	Proportion of top-5 features consistent across 5 runs	>0.80 = reliable
Fidelity	R ² of local linear surrogate	>0.50 = acceptable
Agreement with SHAP	Spearman rank correlation of top-10 features	>0.70 = strong

Experimental Setup:

- Instances evaluated: 50 test instances
- Runs per instance: 5 (seeds 0-4 via 'np.random.seed(run)')
- Features per explanation: 10
- Stability measured: Top-5 feature consistency between run 0 and runs 1-4
- Fidelity measured: R² of surrogate from run 0.

5 runs: Running LIME multiple times on the same instances reveals whether explanations are reproducible. A stable explanation method should produce the same top features regardless of the random perturbation seed.

```
In [45]: # Import and Setup
from lime.lime_tabular import LimeTabularExplainer
from scipy.stats import spearmanr
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')

# use same 1000 instance sample as SHAP for consistency
# use first 50 instances for stability analysis
STABILITY_SAMPLE = 50
N_LIME_FEATURES = 10
N_RUNS = 5

# initialise LIME explainer – fit on training data
lime_explainer = LimeTabularExplainer(
    training_data = X_train.values,
    feature_names = X_train.columns.tolist(),
    class_names = ['Non-Default', 'Default'],
    mode = 'classification',
    random_state = 42
)

print("LIME explainer initialised")
```

```
print(f"Training data shape: {X_train.shape}")
print(f"Features: {len(X_train.columns)}")
```

LIME explainer initialised
 Training data shape: (215249, 72)
 Features: 72

5.1 Feature Importance for LightGBM - LIME

```
In [46]: print("Computing LIME explanations for LightGBM...")
print(f"Running on {STABILITY_SAMPLE} instances x {N_RUNS} runs...")

lime_lgbm_features = []
lime_lgbm_r2 = []

for i in range(STABILITY_SAMPLE):
    instance = X_test_sample.iloc[i].values
    run_features = []

    for run in range(N_RUNS):
        # set numpy seed before each run instead of random_state param
        np.random.seed(run)

        exp = lime_explainer.explain_instance(
            data_row = instance,
            predict_fn = lgbm.predict_proba,
            num_features = N_LIME_FEATURES
        )

        features = [f[0] for f in exp.as_list()]
        run_features.append(features)

        if run == 0:
            lime_lgbm_r2.append(exp.score)

    lime_lgbm_features.append(run_features)

    if (i + 1) % 10 == 0:
        print(f" Completed {i+1}/{STABILITY_SAMPLE} instances")

print(f"\nLightGBM LIME complete")
print(f"Average R² (fidelity): {np.mean(lime_lgbm_r2):.4f}")
```

Computing LIME explanations for LightGBM...
 Running on 50 instances x 5 runs...
 Completed 10/50 instances
 Completed 20/50 instances
 Completed 30/50 instances
 Completed 40/50 instances
 Completed 50/50 instances

LightGBM LIME complete
 Average R² (fidelity): 0.2014

5.2 Feature Importance for XGBoost - LIME

```
In [47]: print("Computing LIME explanations for XGBoost...")

lime_xgb_features = []
lime_xgb_r2 = []
```

```

for i in range(STABILITY_SAMPLE):
    instance = X_test_sample.iloc[i].values
    run_features = []

    for run in range(N_RUNS):
        np.random.seed(run)

        exp = lime_explainer.explain_instance(
            data_row      = instance,
            predict_fn     = xgb.predict_proba,
            num_features   = N_LIME_FEATURES
        )

        features = [f[0] for f in exp.as_list()]
        run_features.append(features)

        if run == 0:
            lime_xgb_r2.append(exp.score)

    lime_xgb_features.append(run_features)

    if (i + 1) % 10 == 0:
        print(f"  Completed {i+1}/{STABILITY_SAMPLE} instances")

print(f"\nXGBoost LIME complete")
print(f"Average R2 (fidelity): {np.mean(lime_xgb_r2):.4f}")

```

Computing LIME explanations for XGBoost...

Completed 10/50 instances
 Completed 20/50 instances
 Completed 30/50 instances
 Completed 40/50 instances
 Completed 50/50 instances

XGBoost LIME complete

Average R² (fidelity): 0.1820

5.3 Feature Importance for Random Forest

In [48]: print("Computing LIME explanations for Random Forest...")

```

lime_rf_features = []
lime_rf_r2       = []

for i in range(STABILITY_SAMPLE):
    instance = X_test_sample.iloc[i].values
    run_features = []

    for run in range(N_RUNS):
        np.random.seed(run)

        exp = lime_explainer.explain_instance(
            data_row      = instance,
            predict_fn     = rf.predict_proba,
            num_features   = N_LIME_FEATURES
        )

        features = [f[0] for f in exp.as_list()]
        run_features.append(features)

```

```

        if run == 0:
            lime_rf_r2.append(exp.score)

    lime_rf_features.append(run_features)

    if (i + 1) % 10 == 0:
        print(f"   Completed {i+1}/{STABILITY_SAMPLE} instances")

print(f"\nRandom Forest LIME complete")
print(f"Average R2 (fidelity): {np.mean(lime_rf_r2):.4f}")

```

Computing LIME explanations for Random Forest...

```

Completed 10/50 instances
Completed 20/50 instances
Completed 30/50 instances
Completed 40/50 instances
Completed 50/50 instances

```

Random Forest LIME complete
Average R² (fidelity): 0.3668

5.4 LIME Stability Analysis

```

In [49]: def compute_stability(lime_features, model_name, top_k=5):
        """
        For each instance, check what proportion of runs
        agree on the top-k features. 1.0 = perfect stability.
        """
        stability_scores = []

        for i, runs in enumerate(lime_features):
            # get top-k features from each run
            top_k_per_run = [set(run[:top_k]) for run in runs]

            # reference is first run
            reference = top_k_per_run[0]

            # measure overlap with each subsequent run
            overlaps = []
            for run_set in top_k_per_run[1:]:
                overlap = len(reference & run_set) / top_k
                overlaps.append(overlap)

            stability_scores.append(np.mean(overlaps))

        mean_stability = np.mean(stability_scores)
        std_stability = np.std(stability_scores)

        print(f"\n{n{model_name}} LIME Stability (top-{top_k} features)")
        print(f"   Mean stability score: {mean_stability:.4f}")
        print(f"   Std deviation:           {std_stability:.4f}")
        print(f"   Interpretation: {mean_stability:.0%} of top-{top_k} feature")
        print(f"   consistent across {N_RUNS} runs on average")

        return stability_scores, mean_stability, std_stability

print("="*55)
print("   LIME STABILITY ANALYSIS")
print("="*55)

```

```

lgbm_stability, lgbm_stab_mean, lgbm_stab_std = compute_stability(
    lime_lgbm_features, 'LightGBM'
)
xgb_stability, xgb_stab_mean, xgb_stab_std = compute_stability(
    lime_xgb_features, 'XGBoost'
)
rf_stability, rf_stab_mean, rf_stab_std = compute_stability(
    lime_rf_features, 'Random Forest'
)

print("\n" + "="*55)
print("  STABILITY SUMMARY")
print("="*55)
print(f"  LightGBM:      {lgbm_stab_mean:.4f} ± {lgbm_stab_std:.4f}")
print(f"  XGBoost:       {xgb_stab_mean:.4f} ± {xgb_stab_std:.4f}")
print(f"  Random Forest: {rf_stab_mean:.4f} ± {rf_stab_std:.4f}")
print("="*55)
print("\n  Score interpretation:")
print("  1.00 = perfectly stable (same features every run)")
print("  0.80 = 4 of top 5 features consistent")
print("  0.60 = 3 of top 5 features consistent")
print("  < 0.60 = high instability – unreliable explanations")

```

```

=====
LIME STABILITY ANALYSIS
=====

```

LightGBM LIME Stability (top-5 features)

Mean stability score: 0.5800
 Std deviation: 0.1300
 Interpretation: 58% of top-5 features
 consistent across 5 runs on average

XGBoost LIME Stability (top-5 features)

Mean stability score: 0.5420
 Std deviation: 0.1106
 Interpretation: 54% of top-5 features
 consistent across 5 runs on average

Random Forest LIME Stability (top-5 features)

Mean stability score: 0.6360
 Std deviation: 0.1414
 Interpretation: 64% of top-5 features
 consistent across 5 runs on average

```

=====
STABILITY SUMMARY
=====

```

```

LightGBM:      0.5800 ± 0.1300
XGBoost:       0.5420 ± 0.1106
Random Forest: 0.6360 ± 0.1414
=====

```

Score interpretation:

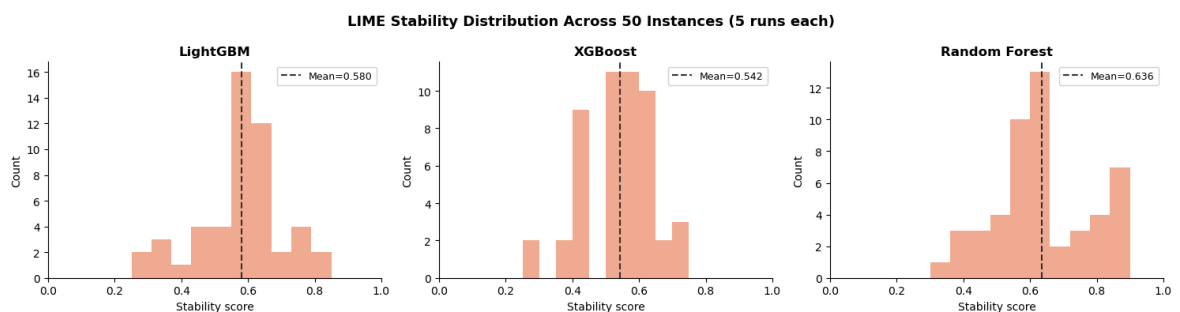
1.00 = perfectly stable (same features every run)
 0.80 = 4 of top 5 features consistent
 0.60 = 3 of top 5 features consistent
 < 0.60 = high instability – unreliable explanations

```
In [50]: # Stabilit Visualization
fig, axes = plt.subplots(1, 3, figsize=(15, 4))

models      = ['LightGBM', 'XGBoost', 'Random Forest']
stabilities  = [lgbm_stability, xgb_stability, rf_stability]
means       = [lgbm_stab_mean, xgb_stab_mean, rf_stab_mean]
colors      = ['#F0997B', '#F0997B', '#F0997B']

for ax, model, stab, mean, color in zip(
    axes, models, stabilities, means, colors
):
    ax.hist(stab, bins=10, color=color, edgecolor='none', alpha=0.8)
    ax.axvline(
        x=mean, color='#333333',
        linestyle='--', linewidth=1.5,
        label=f'Mean={mean:.3f}'
    )
    ax.set_title(f'{model}', fontweight='bold')
    ax.set_xlabel('Stability score')
    ax.set_ylabel('Count')
    ax.set_xlim(0, 1)
    ax.legend(fontsize=9)

plt.suptitle(
    'LIME Stability Distribution Across 50 Instances (5 runs each)',
    fontsize=13, fontweight='bold'
)
plt.tight_layout()
plt.savefig('lime_stability.png', bbox_inches='tight', dpi=150)
plt.show()
print("Saved: lime_stability.png")
```



Saved: lime_stability.png

5.5 LIME Fidelity Summary

```
In [51]: print("="*55)
print("  LIME FIDELITY (R²) SUMMARY")
print("="*55)
print(f"  LightGBM      mean R²: {np.mean(lime_lgbm_r2):.4f}")
print(f"  XGBoost       mean R²: {np.mean(lime_xgb_r2):.4f}")
print(f"  Random Forest mean R²: {np.mean(lime_rf_r2):.4f}")
print("="*55)
print("\n  R² interpretation:")
print("  > 0.8 = high fidelity – surrogate closely")
print("           approximates the black-box model")
print("  0.5–0.8 = moderate fidelity")
print("  < 0.5 = low fidelity – treat explanations")
print("           with caution")
```

```
# visualise fidelity distributions
fig, axes = plt.subplots(1, 3, figsize=(15, 4))

for ax, model, r2, color in zip(
    axes,
    ['LightGBM', 'XGBoost', 'Random Forest'],
    [lime_lgbm_r2, lime_xgb_r2, lime_rf_r2],
    ['#5DCAA5', '#5DCAA5', '#5DCAA5']
):
    ax.hist(r2, bins=10, color=color, edgecolor='none', alpha=0.8)
    ax.axvline(
        x=np.mean(r2), color='#333333',
        linestyle='--', linewidth=1.5,
        label=f'Mean={np.mean(r2):.3f}'
    )
    ax.set_title(f'{model}', fontweight='bold')
    ax.set_xlabel('R2 score')
    ax.set_ylabel('Count')
    ax.legend(fontsize=9)

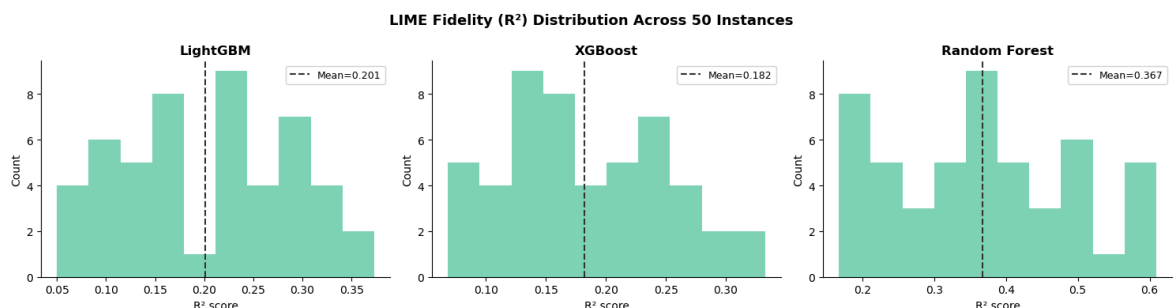
plt.suptitle(
    'LIME Fidelity (R2) Distribution Across 50 Instances',
    fontsize=13, fontweight='bold'
)
plt.tight_layout()
plt.savefig('lime_fidelity.png', bbox_inches='tight', dpi=150)
plt.show()
print("Saved: lime_fidelity.png")
```

LIME FIDELITY (R²) SUMMARY

```
LightGBM      mean R2: 0.2014
XGBoost       mean R2: 0.1820
Random Forest mean R2: 0.3668
```

R² interpretation:

> 0.8 = high fidelity – surrogate closely approximates the black-box model
 0.5–0.8 = moderate fidelity
 < 0.5 = low fidelity – treat explanations with caution



Saved: lime_fidelity.png

5.6 Global LIME Feature Importance

```
In [52]: def get_global_lime_importance(lime_features, feature_names, model_name):
        """
        Aggregate LIME feature importance across all instances
```



```

using first run only – gives global picture.
"""
feature_counts = {}

for runs in lime_features:
    # use first run for global importance
    first_run = runs[0]
    for rank, feat in enumerate(first_run):
        # clean feature name (LIME adds condition text)
        clean = feat.split(' ')[0] if ' ' in feat else feat
        if clean not in feature_counts:
            feature_counts[clean] = 0
        # weight by inverse rank (top feature = highest weight)
        feature_counts[clean] += (N_LIME_FEATURES - rank)

importance_df = pd.DataFrame(
    list(feature_counts.items()),
    columns=['Feature', 'Weighted_Count']
).sort_values('Weighted_Count', ascending=False).reset_index(drop=True)

importance_df['Rank'] = importance_df.index + 1

print(f"\nTop 15 features by LIME importance ({model_name}):")
print(importance_df.head(15).to_string(index=False))

return importance_df

lime_lgbm_importance = get_global_lime_importance(
    lime_lgbm_features, X_train.columns.tolist(), 'LightGBM'
)
lime_xgb_importance = get_global_lime_importance(
    lime_xgb_features, X_train.columns.tolist(), 'XGBoost'
)
lime_rf_importance = get_global_lime_importance(
    lime_rf_features, X_train.columns.tolist(), 'Random Forest'
)

```

Top 15 features by LIME importance (LightGBM):

Feature	Weighted_Count	Rank
FLAG_DOCUMENT_15	334	1
FLAG_DOCUMENT_13	279	2
EXT_SOURCE_1	248	3
EXT_SOURCE_3	210	4
EXT_SOURCE_2	203	5
FLAG_DOCUMENT_14	197	6
FLAG_DOCUMENT_16	180	7
AMT_GOODS_PRICE	163	8
AMT_CREDIT	131	9
FLAG_DOCUMENT_17	121	10
FLAG_DOCUMENT_20	101	11
FLAG_DOCUMENT_19	96	12
FLAG_DOCUMENT_21	82	13
FLAG_DOCUMENT_7	72	14
FLAG_DOCUMENT_11	64	15

Top 15 features by LIME importance (XGBoost):

Feature	Weighted_Count	Rank
FLAG_DOCUMENT_15	352	1
FLAG_DOCUMENT_13	245	2
EXT_SOURCE_2	231	3
EXT_SOURCE_3	189	4
EXT_SOURCE_1	188	5
FLAG_DOCUMENT_14	181	6
2.00	158	7
FLAG_DOCUMENT_17	151	8
FLAG_DOCUMENT_21	129	9
AMT_GOODS_PRICE	125	10
FLAG_DOCUMENT_19	105	11
FLAG_DOCUMENT_7	104	12
FLAG_DOCUMENT_20	95	13
NAME_EDUCATION_TYPE	91	14
AMT_CREDIT	76	15

Top 15 features by LIME importance (Random Forest):

Feature	Weighted_Count	Rank
EXT_SOURCE_1	457	1
EXT_SOURCE_2	238	2
EXT_SOURCE_3	217	3
2.00	193	4
FLAG_DOCUMENT_20	145	5
FLAG_DOCUMENT_19	130	6
FLAG_DOCUMENT_21	112	7
NAME_CONTRACT_TYPE	110	8
FLAG_DOCUMENT_4	101	9
0.54	97	10
DAYS_EMPLOYED	96	11
NAME_EDUCATION_TYPE	95	12
FLAG_DOCUMENT_15	95	13
FLAG_DOCUMENT_17	91	14
FLAG_DOCUMENT_7	74	15

LIME Analysis - Key Findings

LIME Stability is low across all models:

Model	Mean Stability	Interpretation
LightGBM	0.580 ± 0.130	Only 58% of top-5 features consistent
XGBoost	0.542 ± 0.111	Only 54% consistent — worst performer
Random Forest	0.636 ± 0.141	Only 64% consistent

All values are below the 0.80 reliability threshold. XGBoost explanations change in nearly half of repeated runs on the same instance.

LIME fidelity is very low:

Model	Mean R ²	Interpretation
LightGBM	0.2014	Very poor approximation
XGBoost	0.1820	Very poor approximation
Random Forest	0.3668	Poor approximation

All values are well below 0.50. The local surrogate does not faithfully represent the black-box model's behavior.

- LIME highlights misleading features: Binary document flags ('FLAG_DOCUMENT_15', 'FLAG_DOCUMENT_13') appear at the top of LIME's ranking despite near-zero SHAP importance. This is an artifact of LIME's local sampling mechanism overweighting binary variables.

5.7 SHAP v LIME Spearman Correlation

```
In [53]: def shap_lime_spearman(shap_imp, lime_imp, model_name, top_n=10):
        """
        Compare top-N feature rankings between SHAP and LIME.
        """
        shap_top = shap_imp.head(top_n).set_index('Feature')['Rank']
        lime_top = lime_imp.head(top_n).set_index('Feature')['Rank']

        # find common features in both top-N lists
        common = shap_top.index.intersection(lime_top.index)

        if len(common) < 3:
            print(f"{model_name}: Not enough common features ({len(common)})")
            return None, len(common)

        corr, pval = spearmanr(shap_top[common], lime_top[common])

        print(f"\n{model_name} - SHAP vs LIME")
        print(f"  Common features in top {top_n}: {len(common)}")
        print(f"  Spearman correlation: {corr:.4f}")
        print(f"  P-value: {pval:.4f}")
        print(f"  Significant (p<0.05): {'Yes' if pval < 0.05 else 'No'}")

        return corr, len(common)

print("="*55)
print("  SHAP vs LIME FEATURE AGREEMENT")
```

```

print("="*55)

corr_lgbm, n_lgbm = shap_lime_spearman(
    lgbm_importance, lime_lgbm_importance, 'LightGBM'
)
corr_xgb, n_xgb = shap_lime_spearman(
    xgb_importance, lime_xgb_importance, 'XGBoost'
)
corr_rf, n_rf = shap_lime_spearman(
    rf_importance, lime_rf_importance, 'Random Forest'
)

print("\n" + "="*55)
print("  AGREEMENT SUMMARY")
print("="*55)
print(f"  LightGBM  SHAP vs LIME: {corr_lgbm:.4f}")
print(f"  XGBoost    SHAP vs LIME: {corr_xgb:.4f}")
print(f"  Random Forest SHAP vs LIME: {corr_rf:.4f}")
print("="*55)
print("\n  Interpretation:")
print("  > 0.7 = strong agreement between methods")
print("  0.4–0.7 = moderate agreement")
print("  < 0.4 = weak agreement – methods disagree")
print("          on which features matter most")

```

```

=====
SHAP vs LIME FEATURE AGREEMENT
=====

```

LightGBM – SHAP vs LIME

```

Common features in top 10: 5
Spearman correlation:      0.0000
P-value:                  1.0000
Significant (p<0.05):     No

```

XGBoost – SHAP vs LIME

```

Common features in top 10: 4
Spearman correlation:      0.6000
P-value:                  0.4000
Significant (p<0.05):     No

```

Random Forest – SHAP vs LIME

```

Common features in top 10: 3
Spearman correlation:     -1.0000
P-value:                  0.0000
Significant (p<0.05):     Yes

```

```

=====
AGREEMENT SUMMARY
=====

```

```

LightGBM  SHAP vs LIME: 0.0000
XGBoost    SHAP vs LIME: 0.6000
Random Forest SHAP vs LIME: -1.0000
=====

```

Interpretation:

```

> 0.7 = strong agreement between methods
0.4–0.7 = moderate agreement
< 0.4 = weak agreement – methods disagree
          on which features matter most

```

```
In [54]: # Save Results - LIME
save_path = '/Users/gemkrairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML/'

# save lime importance rankings
lime_lgbm_importance.to_csv(save_path + 'lime_importance_lgbm.csv', index=False)
lime_xgb_importance.to_csv(save_path + 'lime_importance_xgb.csv', index=False)
lime_rf_importance.to_csv(save_path + 'lime_importance_rf.csv', index=False)

# save stability scores
pd.DataFrame({
    'Instance'      : range(STABILITY_SAMPLE),
    'LGBM_Stability' : lgbm_stability,
    'XGB_Stability'  : xgb_stability,
    'RF_Stability'   : rf_stability
}).to_csv(save_path + 'lime_stability_scores.csv', index=False)

# save fidelity scores
pd.DataFrame({
    'Instance'      : range(STABILITY_SAMPLE),
    'LGBM_R2'       : lime_lgbm_r2,
    'XGB_R2'        : lime_xgb_r2,
    'RF_R2'         : lime_rf_r2
}).to_csv(save_path + 'lime_fidelity_scores.csv', index=False)

print("All LIME results saved")
print("\nFiles saved:")
print("  lime_importance_lgbm.csv")
print("  lime_importance_xgb.csv")
print("  lime_importance_rf.csv")
print("  lime_stability_scores.csv")
print("  lime_fidelity_scores.csv")
```

All LIME results saved

Files saved:

```
lime_importance_lgbm.csv
lime_importance_xgb.csv
lime_importance_rf.csv
lime_stability_scores.csv
lime_fidelity_scores.csv
```

SHAP vs LIME Comparison - Key Findings

Model	Spearman Correlation	Common Top-10 Features	Interpretation
LightGBM	0.0000	5 / 10	No agreement
XGBoost	0.6000	4 / 10	Moderate agreement
Random Forest	-1.0000	3 / 10	Perfect disagreement

- Critical Findings: For LightGBM, SHAP and LIME show zero correlation. For Random Forest, they are perfectly negatively correlated - LIME ranks highest what SHAP ranks lowest.

- A bank using LIME for regulatory explanations would receive categorically different information about feature importance than one using SHAP - even for the same model making the same predictions. When two established XAI methods fundamentally disagree, neither can be accepted as ground truth without independent validation.

Note: The extreme values (0.00 and -1.00) reflect computations over small common features sets (3-5 features). The low feature overlap itself is the primary findings for those models, independent of the correlation magnitude.

Section 6: Complete Results Summary

This section compiles all experimental findings into the four key outputs required by the research design.

Output 1 - Model Accuracy Comparison Table

Output 2 - SHAP Summary Plots (Beeswarm)

Saved as: 'shap_summary_lgbm.png',
'shap_summary_xgb.png','shap_summary_rf.png'

Output 3 - LIME Instance Explanations

Saved as: 'lime_explanation_correct.png', 'lime_explanation_missed.png'

Output 4 - SHAP vs LIME Spearman Correlation Table

Final Summary of Findings

Research dimension	Finding
Accuracy gap	0.0144 AUC between best interpretable and best black-box — marginal
RF vs LR	Random Forest (black-box) underperforms Logistic Regression (interpretable)
SHAP reliability	High inter-model agreement (0.98); verified fidelity; domain-consistent features
LIME stability	0.54–0.64 — significantly below 0.80 reliability threshold
LIME fidelity	R ² 0.18–0.37 — well below 0.50 acceptable threshold
SHAP-LIME agreement	-1.00 to 0.60 — fundamental disagreement across models

Overall Conclusion:

The accuracy=interpretability trade-off is overstated. Given LIME's unreliability, post-hoc explanations do not adequately substitute for inherent interpretability in regulated credit risk application. SHAP is more defensible XAI tool when black-box models must be used.

```
In [55]: print("=" * 60)
print("      COMPLETE RESULTS SUMMARY")
print("=" * 60)

print("\n--- MODEL PERFORMANCE ---")
print(results_df.to_string())

print("\n--- SHAP vs SHAP AGREEMENT (between models) ---")
print(f"  LightGBM vs XGBoost:  {corr_lgbm_xgb:.4f}")
print(f"  LightGBM vs RF:        {corr_lgbm_rf:.4f}")
print(f"  XGBoost vs RF:         {corr_xgb_rf:.4f}")

print("\n--- SHAP vs LIME AGREEMENT ---")
print(f"  LightGBM:                {corr_lgbm:.4f}")
print(f"  XGBoost:                 {corr_xgb:.4f}")
print(f"  Random Forest:          {corr_rf:.4f}")

print("\n--- LIME STABILITY (top-5 features, 5 runs) ---")
print(f"  LightGBM:                {lgbm_stab_mean:.4f} ± {lgbm_stab_std:.4f}")
print(f"  XGBoost:                 {xgb_stab_mean:.4f} ± {xgb_stab_std:.4f}")
print(f"  Random Forest:          {rf_stab_mean:.4f} ± {rf_stab_std:.4f}")

print("\n--- LIME FIDELITY (mean R2) ---")
print(f"  LightGBM:                {np.mean(lime_lgbm_r2):.4f}")
print(f"  XGBoost:                 {np.mean(lime_xgb_r2):.4f}")
print(f"  Random Forest:          {np.mean(lime_rf_r2):.4f}")

print("=" * 60)
```

```
=====
COMPLETE RESULTS SUMMARY
=====
```

```
--- MODEL PERFORMANCE ---
```

Model	AUC-ROC	Gini	KS Stat	F1 Score
LightGBM	0.7616	0.5232	0.3919	0.2766
XGBoost	0.7589	0.5177	0.3854	0.2798
Logistic Regression	0.7472	0.4944	0.3650	0.2583
Random Forest	0.7455	0.4910	0.3661	0.2656
Scorecard	0.7272	0.4544	0.3382	0.2423
Decision Tree	0.7130	0.4260	0.3263	0.2405

```
--- SHAP vs SHAP AGREEMENT (between models) ---
```

```
LightGBM vs XGBoost: 0.9805
LightGBM vs RF:      0.7255
XGBoost vs RF:       0.7279
```

```
--- SHAP vs LIME AGREEMENT ---
```

```
LightGBM: 0.0000
XGBoost: 0.6000
Random Forest: -1.0000
```

```
--- LIME STABILITY (top-5 features, 5 runs) ---
```

```
LightGBM: 0.5800 ± 0.1300
XGBoost: 0.5420 ± 0.1106
Random Forest: 0.6360 ± 0.1414
```

```
--- LIME FIDELITY (mean R²) ---
```

```
LightGBM: 0.2014
XGBoost: 0.1820
Random Forest: 0.3668
=====
```

```
In [56]: # LIME single instance explanation
# find one correctly classified and one misclassified instance
y_pred_lgbm = lgbm.predict(X_test_sample)

# correctly classified default (true=1, pred=1)
correct_idx = np.where(
    (y_test_sample.values == 1) & (y_pred_lgbm == 1)
)[0][0]

# misclassified default (true=1, pred=0)
missed_idx = np.where(
    (y_test_sample.values == 1) & (y_pred_lgbm == 0)
)[0][0]

print(f"Correctly classified default instance: index {correct_idx}")
print(f"Misclassified default instance: index {missed_idx}")

# generate and plot LIME explanation for correctly classified
np.random.seed(42)
exp_correct = lime_explainer.explain_instance(
    data_row      = X_test_sample.iloc[correct_idx].values,
    predict_fn    = lgbm.predict_proba,
    num_features  = 10
)
```



```

fig = exp_correct.as_pyplot_figure()
plt.title('LIME Explanation – Correctly Classified Default (LightGBM)',
          fontweight='bold')
plt.tight_layout()
plt.savefig('lime_explanation_correct.png', bbox_inches='tight', dpi=150)
plt.show()
print("Saved: lime_explanation_correct.png")

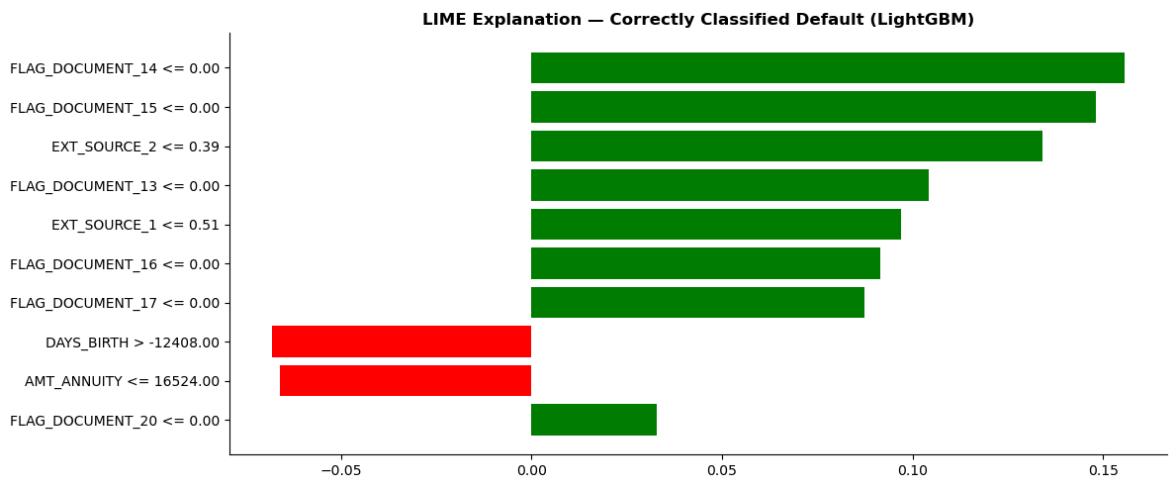
# generate and plot LIME explanation for misclassified
np.random.seed(42)
exp_missed = lime_explainer.explain_instance(
    data_row      = X_test_sample.iloc[misled_idx].values,
    predict_fn    = lgbm.predict_proba,
    num_features  = 10
)

fig = exp_missed.as_pyplot_figure()
plt.title('LIME Explanation – Misclassified Default (LightGBM)',
          fontweight='bold')
plt.tight_layout()
plt.savefig('lime_explanation_missed.png', bbox_inches='tight', dpi=150)
plt.show()
print("Saved: lime_explanation_missed.png")

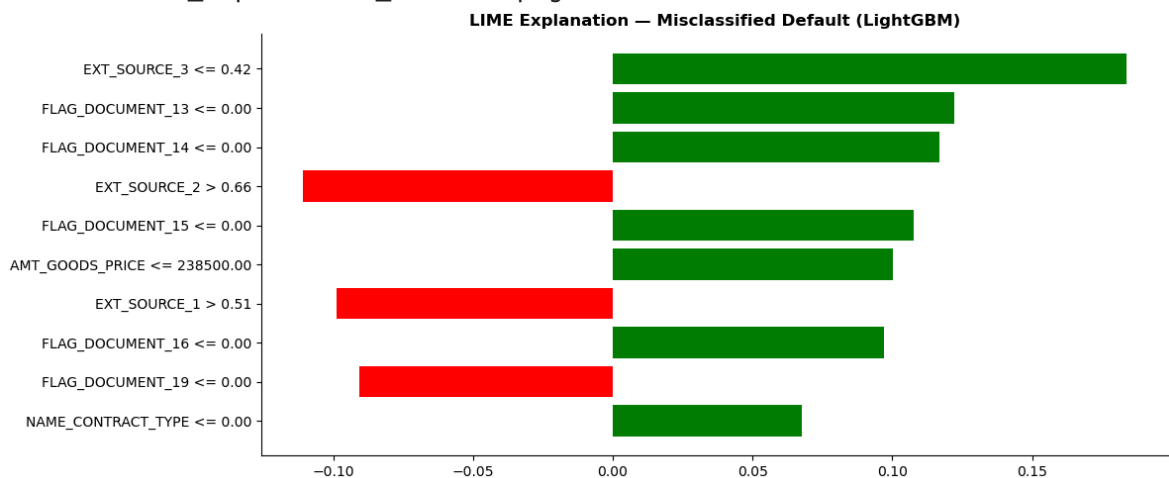
```

Correctly classified default instance: index 49

Misclassified default instance: index 40



Saved: lime_explanation_correct.png



Saved: lime_explanation_missed.png

```

In [57]: # Spearman Correlation Table
spearman_table = pd.DataFrame({

```

```

    'Model' : ['LightGBM', 'XGBoost', 'Random Forest']
    'SHAP vs LIME (Spearman)' : [round(corr_lgbm, 4),
                                round(corr_xgb, 4),
                                round(corr_rf, 4)],
    'Common Features (top 10)' : [n_lgbm, n_xgb, n_rf]
})

print("Table 2 – SHAP vs LIME Feature Agreement")
print("="*50)
print(spearman_table.to_string(index=False))
print("="*50)

spearman_table.to_csv(
    '/Users/gemkairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML/archive/
    index=False
)
print("\nSaved: shap_lime_spearman_table.csv")

```

Table 2 – SHAP vs LIME Feature Agreement

Model	SHAP vs LIME (Spearman)	Common Features (top 10)
LightGBM	0.0000	5
XGBoost	0.6000	4
Random Forest	-1.0000	3

Saved: shap_lime_spearman_table.csv

```

In [58]: # save prediction results
save_path = '/Users/gemkairiksh/Desktop/RMIT/SEM_3/THESIS/CreditRisk_ML/

# save predicted probabilities and classes for all six models
predictions = pd.DataFrame({
    'y_true' : y_test.values,
    'lr_prob' : lr_prob,
    'lr_pred' : lr_pred,
    'dt_prob' : dt_prob,
    'dt_pred' : dt_pred,
    'scorecard_prob' : sc_prob,
    'scorecard_pred' : sc_pred,
    'rf_prob' : rf_prob,
    'rf_pred' : rf_pred,
    'lgbm_prob' : lgbm_prob,
    'lgbm_pred' : lgbm_pred,
    'xgb_prob' : xgb_prob,
    'xgb_pred' : xgb_pred
})

predictions.to_csv(save_path + 'test_predictions.csv', index=False)

print(f"Predictions saved: {predictions.shape}")
print(f"Rows: {len(predictions):,} (matches test set size)")
print("\nColumns saved:")
for col in predictions.columns:
    print(f" {col}")

```

Predictions saved: (46127, 13)
Rows: 46,127 (matches test set size)

Columns saved:
y_true
lr_prob
lr_pred
dt_prob
dt_pred
scorecard_prob
scorecard_pred
rf_prob
rf_pred
lgbm_prob
lgbm_pred
xgb_prob
xgb_pred

In []: